

tiptotyp

The contributor induction

Every surface. Every boundary.
Every place to start.

129 SLIDES / 10 CHAPTERS / SOURCE LINKS THROUGHOUT



Why this app exists

A Typst-first desktop workbench, not merely a text box next to a PDF.

- Purpose and product boundaries
- Evidence and reading routes
- The invariants worth defending

The job: make local Typst work feel direct

Write the source, understand the document, inspect the result, keep control of the files.

01 / WRITE

Edit Typst in a native Rust application. Keep source and preview together, or give either one the window. Browse the project without leaving the authoring loop.

02 / UNDERSTAND

Use language-server help and local structural analysis: completion, diagnostics, hover information, headings, definitions, imports, labels and references.

03 / TRUST

Treat exported PDF bytes as a canonical compiler artifact. The interactive frontend is for feedback; a display transform or preview fallback must not rewrite the document.

04 / OWN

Work with ordinary local files and explicit saving. Private preview material belongs in .tip-toptyp, not in the user's manuscript or staged Git changes.

Read this as a contributor's field manual

Inventory first; ownership second; safe changes third.

WHAT WAS INSPECTED

Public master source and repository documentation, plus official competitor documentation, inspected on 20 September 2026. Links point to branch files, not an immutable commit.

WHAT WAS NOT VERIFIED

This is a static source audit. The application was not launched, its Rust tests were not executed here, and platform behavior was not independently reproduced.

HOW CLAIMS ARE LABELLED

SOURCE AUDIT means implemented in inspected code or documented by the project. COMPARISON means publisher documentation. PROPOSAL means a recommended change, not a shipped feature.

HOW TO NAVIGATE

Use the PDF outline and the contents page. Feature slides lead into a UI census, architecture, implementation contracts, feature-to-file routes, contribution checks and a staged roadmap.

Pick your route through the codebase

Read straight through, or jump to the chapter that matches the change you are making.

01	Why this app exists	2	06	The contracts below the UI	57
02	What contributors are maintaining	7	07	"I want to change X. Where is Y?"	82
03	Compare without inventing exclusivity	28	08	How to contribute without guessing	94
04	The UI census	35	09	The exact registries	102
05	The architecture from orbit	47	10	What should get better—and how	113

Four invariants to carry into every patch

A useful change preserves the model, not just the screenshot.

SOURCE IS THE AUTHORITY

An editor projection is not the saved document. Canonical Typst must feed saving, compilation and the language server; display text may have another coordinate system.

OWNERSHIP IS EXPLICIT

Window identity, document identity, document revision and service generation are different things. A result is useful only to the owner and version that requested it.

EFFECTS HAVE A LIFETIME

A replaceable search or scan may discard stale results. A save or Git mutation cannot be wished away because its window disappeared.

UI IS NOT THE DESKTOP

egui content, native child viewports and the embedded webview occupy different rendering/focus systems. A framebuffer capture cannot prove the whole composed desktop is correct.

SOURCE `src/mitex_document.rs · core/src/document.rs · src/worker/exclusive.rs · docs/ui-qa-screenshots.md`

What contributors are maintaining

A feature inventory that includes the easily missed parts.

- Documents, editing and navigation
- Preview, Git, packages and appearance
- Platform limits and deliberate non-features

Documents and project entry points

Four document kinds; several ways into a workspace.

SURFACE	IMPLEMENTED BEHAVIOR	IMPORTANT BOUNDARY
Typst	First-class authoring, language services and compilation.	Only Typst has the full compile/preview pipeline.
Other text	Open and edit UTF-8 project files with general syntax support.	Editing .tex or .md is not a LaTeX/Markdown build system.
PDF	Read a PDF as a document tab.	Binary content is not an editable text buffer.
Images	Display common raster formats as document tabs.	Content signatures precede misleading extensions.
Open / workspace	Open files, directories, command-line paths and native open requests.	Root discovery uses nearby project markers; root can be changed.
Remembered workspace	Recent roots and the last opened file are persisted.	This is not full restoration of all tabs and unsaved buffers.

SOURCE [src/document.rs](#) · [src/launch.rs](#) · [src/open_requests.rs](#) · [src/settings.rs](#)

Tabs are real document state, not labels

Active source and the document driving preview are distinct choices.

WORK WITH SEVERAL FILES

Open, select, reorder and close tabs; navigate next/previous by action. Dirty-document handling applies to closing as well as replacing a document.

KEEP YOUR PLACE

A tab carries its document, folding state and TextEdit state. Switching tabs should not throw away cursor/selection, undo ownership or view context.

CHOOSE THE PREVIEW ENTRY

A separate preview-tab identity lets a main document remain the output target while a supporting file is active. An explicit choice must survive ordinary tab selection.

DO NOT MULTIPLY SERVICES

A TabRecord is not another EditorApp. Window-level services coordinate the document set; opening a tab must not accidentally create a second independent application session.

The everyday source-editing kit

Small features are part of the product contract too.

TEXT OPERATIONS

Undo, redo, cut, copy, paste and select-all; explicit Save and Save As; comment toggling and formatting for the appropriate document type.

READABILITY CONTROLS

Line wrapping, line numbers, syntax colors, current editing context and independent code/UI typography. Code, Split and Preview modes change the workspace emphasis.

DELIMITER ASSISTANCE

Automatic delimiter pairs, matching-delimiter feedback and rainbow brackets. These are separate behaviors, with separate state and tests—not one “brackets” toggle in the implementation.

EDITOR INTERACTION

Context menus, configurable shortcuts, caret/pointer tooltip actions, completion invocation and selection-aware edits. Availability depends on the current document and live service state.

SOURCE `src/auto_pairs.rs` · `src/delimiters.rs` · `src/rainbow.rs` · `src/app/editor_view.rs`

Find and replace: more than literal search

Document-local search has its own revision-aware model.

SEARCH MODES

Literal or regular-expression queries; case sensitivity; next/previous navigation with wraparound. An invalid regex has an explicit error state rather than looking like zero matches.

UNICODE IS A CONTRACT

Matches store UTF-8 byte ranges and Unicode-scalar cursor ranges. Case-insensitive matching maps results back to original source boundaries.

REPLACE SAFELY

Replace one or all matches. Replacement text is literal even in regex mode: capture substitutions such as \$1 are not a promised feature.

KNOW THE SCOPE

This is search in the current source document. Explorer filtering is a different feature; repository-wide file-content search is a future improvement, not a hidden synonym.

SOURCE `src/search.rs · src/app/find_bar.rs · src/app/explorer_view.rs`

Language intelligence and completion

Tinymist is the semantic engine; the application owns safe application of results.

SEMANTIC HELP

Diagnostics, hover descriptions, completion suggestions and document formatting arrive through the language-server connection. Source/preview navigation uses the interactive preview path.

LOCAL PRESENTATION

Fuzzy filtering, human-readable detail and reference-aware labels make server results usable. Font completions add local catalog knowledge.

EDIT TRANSACTIONS

A selected completion can contain a main edit plus additional edits. Validate and apply them together, producing one undo operation rather than partial mutations.

SNIPPET BOUNDARY

Snippet syntax is expanded and an insertion cursor is chosen. Do not advertise a full persistent multi-tabstop snippet session merely because snippet expansion exists.

SOURCE `src/tinymist/protocol.rs · src/completion.rs · src/completion_edit.rs · src/app/completion_popup.rs`

Structure in the editor itself

Navigation is not confined to the sidebar.

STICKY CONTEXT

Heading ancestry and lexical context stay visible near the top of the editor. Context includes bindings, functions and blocks as well as document headings.

FOLDING

Toggle a fold, collapse all or expand all. Hidden rows retain source positions; navigation reveals requested locations without changing the document text.

TABLE / GRID EDITOR

A dedicated editor handles conservative rectangular table/grid forms. Add/remove rows or columns and apply a validated source replacement.

CONSERVATIVE BY DESIGN

Dynamic cells, computed structure and unsupported syntax are refused rather than flattened. A small reliable structure editor is not a general WYSIWYG promise.

SOURCE [src/editor_features.rs](#) · [src/folding.rs](#) · [src/app/native_views.rs](#)

miTeX dollar mode: an unusual differentiator

A controlled TeX-math editing projection over canonical Typst.

WHAT THE USER SEES

A dollar-math-oriented editing view for supported miTeX calls, with a configurable mode and local TeX completion support.

WHAT THE FILE CONTAINS

Canonical Typst remains the authoritative representation. Saving, compiler requests and LSP synchronization must not serialize the projected display buffer.

WHAT GETS CHECKED

Projection creation validates syntax and the renderer import/binding. Edits require a reversible mapping and must not split Unicode boundaries.

WHAT IT IS NOT

It is not a TeX engine, arbitrary LaTeX conversion, or a visual editor for every Typst construct. Unsupported or ambiguous source should fail closed.

SOURCE `src/mitex_projection.rs · src/mitex_document.rs · src/app/mitex_mode.rs · src/tex_completion.rs`

Explorer: the project as several indexes

A filesystem tree and a semantic navigator share one sidebar.

FILES AND GIT

Browse the workspace tree and repository state. Open files and invoke context actions without treating every filesystem object as source text.

CONTENTS AND SUBFILES

Navigate headings and literal local import/include relationships. Clicking an indexed item should identify the correct document and position.

SYMBOLS AND PACKAGES

Find definitions/functions and imported packages. These are structural views, not an evaluator of arbitrary Typst code.

TAGS AND REFERENCES

See declared labels and reference occurrences. Reorder and resize Explorer sections; filter relevant items while retaining stable tree identities.

SOURCE `src/explorer.rs · src/app/explorer_view.rs · src/project_index.rs`

Filesystem management: useful, but bounded

Do not conflate a competent explorer with a complete file manager.

PROJECT OPERATIONS

Change or refresh the workspace, create supported files/folders, rename or delete through application workflows, and handle external drops.

SAFETY RULES

Resolve paths against the workspace boundary. Private artifacts are excluded. A symlink displayed as a leaf must not become unbounded recursive traversal.

WHAT SEARCH DOES

Explorer search filters names, paths and indexed structure. Matching ancestors are revealed; scan generation must not reset tree openness or selection.

WHAT IS STILL MISSING

The current tree explicitly disables multi-selection. General in-tree move/drag reorganization and cross-file content search need dedicated designs and tests.

SOURCE `src/app/workspace_view.rs · src/app/explorer_view.rs · src/workspace.rs`

Live preview: the interactive route

Tinymist serves the frontend; the application embeds and coordinates it.

IMMEDIATE FEEDBACK

An interactive vector preview accompanies source editing on supported platforms. The app consumes a version-matched Tinymist frontend rather than reimplementing its private diff protocol.

BIDIRECTIONAL CONTEXT

The interactive route supports source/preview navigation. Gesture and command choices belong to application policy; document mapping still has to reject stale positions.

PAUSE WITHOUT GOING BLIND

Automatic preview refresh can be paused while editor language services continue receiving changes. Manual compilation remains a separate action.

RECOVER VISIBLY

Failures pass through a bounded recovery policy before the raster route is selected. Status must explain degradation rather than leave a blank pane with a healthy-looking badge.

SOURCE `src/tinymist.rs · src/tinymist_sync.rs · src/preview.rs · core/src/recovery.rs`

PDF generation and export: a separate promise

The exported artifact is not a screenshot of whichever preview happened to be visible.

CANONICAL PDF

A Typst compiler process produces actual PDF bytes. The resulting artifact is versioned and immutable once published to consumers.

COMPILE VS EXPORT

Compile targets the usual PDF destination for the document; Export PDF chooses another destination. Unsaved documents require destination handling.

FRESHNESS

Reuse an artifact only when its key matches the requested source/build context.
Dependency-triggered builds can change output without changing the source buffer revision.

DISPLAY INDEPENDENCE

Rasterization, preview inversion and page residency do not alter exported bytes. A failed preview renderer should not invalidate a successfully produced canonical PDF.

SOURCE `src/compiler.rs · src/app/saves.rs · core/src/preview.rs · src/app/raster_view.rs`

Raster preview: the recovery path is a real reader

It is not the same feature set as the interactive frontend.

DOCUMENT NAVIGATION

Continuous pages, zoom controls, page navigation and fit-to-width behavior. Scroll and pointer-anchor handling are part of the reader experience.

DARK READING

A presentation-only color transform can make pages comfortable against dark chrome without modifying the original pixels/artifact on disk.

PDF TOOLING

Poppler tools inspect, rasterize and extract links: `pdfinfo`, `pdftoppm` and `pdftohtml` have distinct roles. Missing optional capabilities should be identified separately.

LIMITS

Raster pages do not inherit interactive source spans. Linux currently uses the raster route; macOS/Windows additionally support the embedded interactive route.

SOURCE `src/pdf.rs` · `src/pdf_pages.rs` · `src/capabilities.rs` · `src/preview.rs`

Assets, fonts and useful inspection tools

Not every popup is a menu; not every font affects the compiled document.

ASSET HOVER

Preview supported literal image/PDF targets and Explorer assets. Loading, success and failure are explicit states, and requests must remain tied to the hovered target.

DOCUMENT FONTS

Font discovery and previews support document font selection/completion. Literal source edits and compiler font inputs are different from application UI font preferences.

UI AND CODE FONTS

Choose interface and code fonts independently, including weight-related settings. Font loading must propagate to the viewport that actually renders the text.

UNICODE ASSISTANCE

Unicode/font preview infrastructure supports richer completion and glyph rendering. Keep fallback-font and preview-cache behavior out of per-frame filesystem work.

SOURCE [src/asset.rs](#) · [src/font_catalog.rs](#) · [src/font_preview.rs](#) · [src/unicode_fonts.rs](#)

Git: repository workflow inside the authoring loop

This is substantially more than a branch name in the status bar.

REPOSITORY OPERATIONS

Inspect status, stage or unstage files, commit, browse history, fetch, fast-forward-only pull and push through the project's Git integration.

DIFF PRESENTATION

Review staged and unstaged changes; choose unified or side-by-side presentation. Editor gutter feedback compares against HEAD and includes unsaved buffer edits.

HUNK ACTIONS

Inspect a changed chunk, navigate previous/next, stage or unstage a hunk, or revert it through an undoable editor change.

PROTECT PRIVATE MATERIAL

Prevent staging .tiptotyp artifacts. Already-staged private material must remain discoverable for unstaging rather than disappear from the repair path.

SOURCE [src/git.rs](#) · [src/git/editor.rs](#) · [src/git/repository.rs](#) · [src/git/repository/hunks.rs](#)

Package browser: inventory, not magical installation

Local package knowledge survives a registry outage.

DISCOVERY

Merge installed package roots with published registry metadata. Distinguish local data/cache roots and custom roots; expose scan warnings.

BROWSING

Search/filter the catalog; inspect available and installed versions, descriptions, website links and update availability.

USE IN SOURCE

Copy a versioned import expression. This is useful package discovery, not proof of a general one-click installer or a private registry publishing workflow.

REMOVE DELIBERATELY

Uninstall a particular local installation through confirmation and background mutation. An unavailable official index should still leave local packages visible.

SOURCE `src/package_catalog.rs · src/app/package_browser.rs · src/app/native_views.rs`

Appearance: a surprisingly deep product area

Theme, syntax, document page and native chrome are related but separate layers.

BUILT-IN CHOICE

The repository documents 32 bundled themes across multiple families. Light and dark selections are independent rather than one hard-coded palette pair.

IMPORT AND TRANSFORM

Import compatible TextMate/Sublime color schemes. Apply inversion and hue-shift transforms consistently instead of recoloring one surface in isolation.

FINE CONTROL

Typst syntax overrides, page appearance, UI scaling, fonts/weights, title-bar menus and fixed-width-tab preferences provide several levels of customization.

WINDOW IDENTITY

Window-logo/color controls and native chrome matter in multi-window work. Appearance changes must reach child viewports and the embedded preview, not only the root frame.

SOURCE [src/syntax_theme.rs](#) · [src/settings.rs](#) · [src/window_logo.rs](#) · [src/presentation.rs](#)

Settings and shortcuts are first-class interfaces

A contributor can break discoverability without breaking compilation.

FIND THE PREFERENCE

Settings has search/jump targets and a separate viewport. Tool paths, backend state and project-root information are exposed rather than hidden in a configuration file.

REMAP ACTIONS

The shortcut editor searches actions/groups, captures chords, disables assignments and resets one/all bindings. Stable action IDs keep persistence independent of labels.

HANDLE COLLISIONS

Effective shortcuts normalize conflicts and modifier specificity. Native menu equivalents and in-app routing must agree about which action owns the chord.

EXPLAIN DISABILITY

Unavailable actions should be disabled for a concrete reason: document type, missing tool or service state. A single generic “broken” indicator is inadequate.

SOURCE `src/app/settings_panel.rs · src/app/settings_window.rs · src/app/shortcut_editor.rs · src/shortcuts.rs`

Saving, closing and conflict handling

Reliability features are user-facing even when the happy path hides them.

EXPLICIT AND AUTOMATIC SAVE

Save/Save As coexist with configurable autosave timing. A save captures immutable canonical source rather than borrowing a buffer that will continue changing.

EXTERNAL MODIFICATION

Disk fingerprints are revalidated inside the protected write path. A conflicting change leads to an overwrite decision instead of silently treating the original read as current.

CLOSE AND QUIT

Unsaved-document decisions and pending mutations participate in the close workflow. A successful save of an older revision does not authorize discarding a newer one.

LATE OUTCOMES

Operations that finish after the initiating window closes still produce meaningful status. "The owner vanished" is not an acceptable loss of the write result.

SOURCE `src/save_transaction.rs · src/save_io.rs · src/workflow.rs · src/windowing.rs`

Desktop integration and operational visibility

The shell is an application feature, not incidental scaffolding.

MULTI-WINDOW

Create a new document window or open into another window. The shell routes native commands to the focused owner and coordinates application-wide settings.

PLATFORM LIFECYCLE

Native open requests, macOS dock/Finder behavior, title-bar dragging, minimize/fullscreen and close semantics interact with the GUI event loop.

TOOL VISIBILITY

Configured/bundled tools, fallback reasons and capability state are visible. Build identification is available through `-version` and the application's diagnostics surfaces.

SUPPORT MATERIAL

Status history, UI capture scenes, tracing and profiling workflows help contributors turn "it feels wrong" into a reproducible report.

SOURCE `src/windowing.rs` · `src/native_window.rs` · `src/build_info.rs` · `src/screenshot.rs`

Feature boundaries: what not to promise

“Not shipped” and “not the product’s chosen scope” are different judgments.

AREA	CURRENT BOUNDARY	CONTRIBUTOR CONSEQUENCE
Other languages	Text editing is broader than Typst compilation.	Do not imply .tex/.md support includes build pipelines.
Large editing	Whole-buffer TextEdit remains the editor foundation.	Rope storage and viewport-only layout need staged engineering.
Sessions	MRU roots/last file, not complete tab recovery.	Restore is a separate persistence/recovery feature.
Research workflow	Bibliography source/reference awareness, not a full manager.	Zotero/library metadata requires new domain state.
Project ergonomics	No general command palette or full file-content search in reviewed scope.	Reuse command/index infrastructure rather than parallel state.
Collaboration / AI	Not established as built-in product capabilities in this audit.	Treat as optional product decisions, not assumed regressions.

SOURCE README.md · src/settings.rs · editor-comparison.typ · src/app/explorer_view.rs

Compare without inventing exclusivity

A feature can be distinctive without being globally unique.

- Official documentation beats branding
- Concrete asymmetries, not blanket victories
- What tiptotyp could borrow

The comparison method

Three editors named in the request, reviewed against their own documentation.

NAME RESOLUTION

The product referred to as “oleafy” is treated here as Oleafly. Typstify and Texpile are separate products; an unrelated similarly named Rust crate is not evidence about Typstify.

EVIDENCE LEVELS

YES means documented or implemented in the reviewed material. ND means not documented there—not proven absent. LIMIT means an explicit documented limitation.

NO SPEED TROPHIES

No comparative latency, memory or reliability measurements were run. “Native Rust” is an implementation description, not a benchmark result.

REPOSITORY NOTES ARE LEADS

editor-comparison.typ is useful context, but current official pages resolve conflicts. Product marketing names such as “Visual” are not specifications.

The user-visible asymmetries that matter

Claims below are bounded by the official pages inspected on the audit date.

CAPABILITY	TIPTOPTYP	COMPARATORS
Typst source ↔ preview	Interactive route supports navigation.	Typstify manual: one-way; Oleafly Typst tool: un-available.
Git beyond gutter marks	Stage/unstage, commit, remotes and hunk ac-tions.	Typstify handbook documents markers/branch/status; fuller actions ND.
miTeX dollar projection	Implemented over canonical Typst.	Equivalent projection ND in reviewed Typstify/Oleafly/Texpile docs.
Detached output window	Not established in current implementation.	Oleafly explicitly documents a live detached PDF window.
Broad language builds	Typst-first; other text remains editable.	Texpile and Oleafly document LaTeX/Typst/Markdown workflows.
Research / agents	No comparable built-in suite established here.	Typstify/Oleafly/Texpile document reference or as-sistant integrations.

SOURCE [src/tinymist.rs](#) · [src/git/repository/hunks.rs](#) · [Typstify](#) · [handbook v1.8](#) · [Oleafly](#) · [Typst editor](#) · [Texpile](#) · [documentation](#)

Typstify: substantial overlap, real differences

Bundled compiler + Tinymist is not unique to tiptotyp.

SHARED GROUND

The handbook documents local Typst authoring, bundled tools with external overrides, completion/hover, tabs, outline navigation and a filesystem explorer.

DOCUMENTED EXTRA REACH

Typstify includes workspace `sys.inputs` configuration, managed Zotero bibliography synchronization, TPIX package workflows and multiple export formats.

TIPTOPTYP'S STRONGER DOCUMENTED EDGE

The reviewed Typstify Git section stops at gutter, branch and repository status. tiptotyp's explicit hunk/index workflow and bounded fallback machinery go further than that documentation.

A SPECIFIC NAVIGATION DIFFERENCE

Typstify's current handbook describes editor-to-preview navigation as one-way. That is a useful contrast; claiming it has "no synchronization" would be false.

SOURCE [Typstify · handbook v1.8](#) · [Typstify · official site](#) · [src/git/repository/hunks.rs](#) · [core/src/recovery.rs](#)

Oleafly: do not be fooled by the word “Visual”

The dedicated Typst tool page explicitly describes a split source-and-preview layout.

CLARIFY THE CLAIM

Its “Visual Typst Editor” tool is not a rich-text Typst editor. That page also explicitly says Typst projects currently lack SyncTeX-style source/PDF navigation.

A STRONGER READER

The PDF documentation includes search, text selection, two-page layout, reading mode, rotation and a detached OS window that follows new builds.

A BROADER WORKSPACE

Official documentation covers research sources, bibliography, optional AI/agents, conversions and ordinary Git workflows. Do not present local files or Git as tiptotyp exclusives.

USEFUL BORROWING

A detached canonical-PDF reader and clearer stale-preview communication are sensible targets. Recreating the entire research/agent platform is a separate product decision.

Texpile: broader editing and collaboration scope

Compare Typst ergonomics against a multi-language authoring application.

WHAT IT DOCUMENTS

Local LaTeX/Typst/Markdown projects, source and supported visual editing, Vim/Emacs keymaps, multiple cursors, and prose-aware spelling/grammar assistance.

COLLABORATION

Comments, version control and live collaboration are documented. Publisher claims about encryption are not an independent security assessment by this audit.

WHAT STILL STANDS OUT HERE

tiptotyp's miTeX projection, explicit multi-window ownership contracts, independent canonical artifacts and native-child QA model are well-evidenced implementation differentiators.

DO NOT OVERCLAIM

Texpile's docs not describing a particular cache, recovery budget or hunk policy does not prove it lacks one. Mark those engineering details ND, not "No".

SOURCE [Texpile · documentation](#) · [Texpile · version control](#) · [src/mitex_document.rs](#) · [src/child_view.rs](#)

Distinctive bundle: where evidence is strongest

These are precise tiptotyp mechanisms; ND is not a proof of competitor absence.

MECHANISM	TIPTOTYP EVIDENCE	COMPARISON CONCLUSION
Two output routes	Interactive frontend + canonical PDF + raster fallback.	More specific than the generic label "live preview".
Bounded recovery	Explicit generation-aware retry state machine.	Equivalent failure policy ND in reviewed competitor docs.
Projection safety	miTeX source/display mapping with refusal paths.	Equivalent dollar-view contract ND across re-viewed docs.
Unsaved Git hunks	Buffer-aware diff and validated index mutation.	Do not reduce this to generic "Git supported".
Native child lifecycle	Four roles, focus policies, owner-scoped identities.	Contributor-facing differentiator, not a visible button.
QA architecture	43 registered scenes; semantic and geometry checks.	A maintainability capability, not a claim of zero bugs.

SOURCE [src/compiler.rs](#) · [core/src/recovery.rs](#) · [src/mitex_projection.rs](#) · [src/git/repository/hunks.rs](#) · [src/child_view.rs](#) · [src/screenshot.rs](#)

The UI census

Count identities and roles—not screenshots and open instances.

- Windows, popups, modals and local overlays
- Eight Explorer sections and settings targets
- Commands, shortcuts and QA scene coverage

The numbers you can actually defend

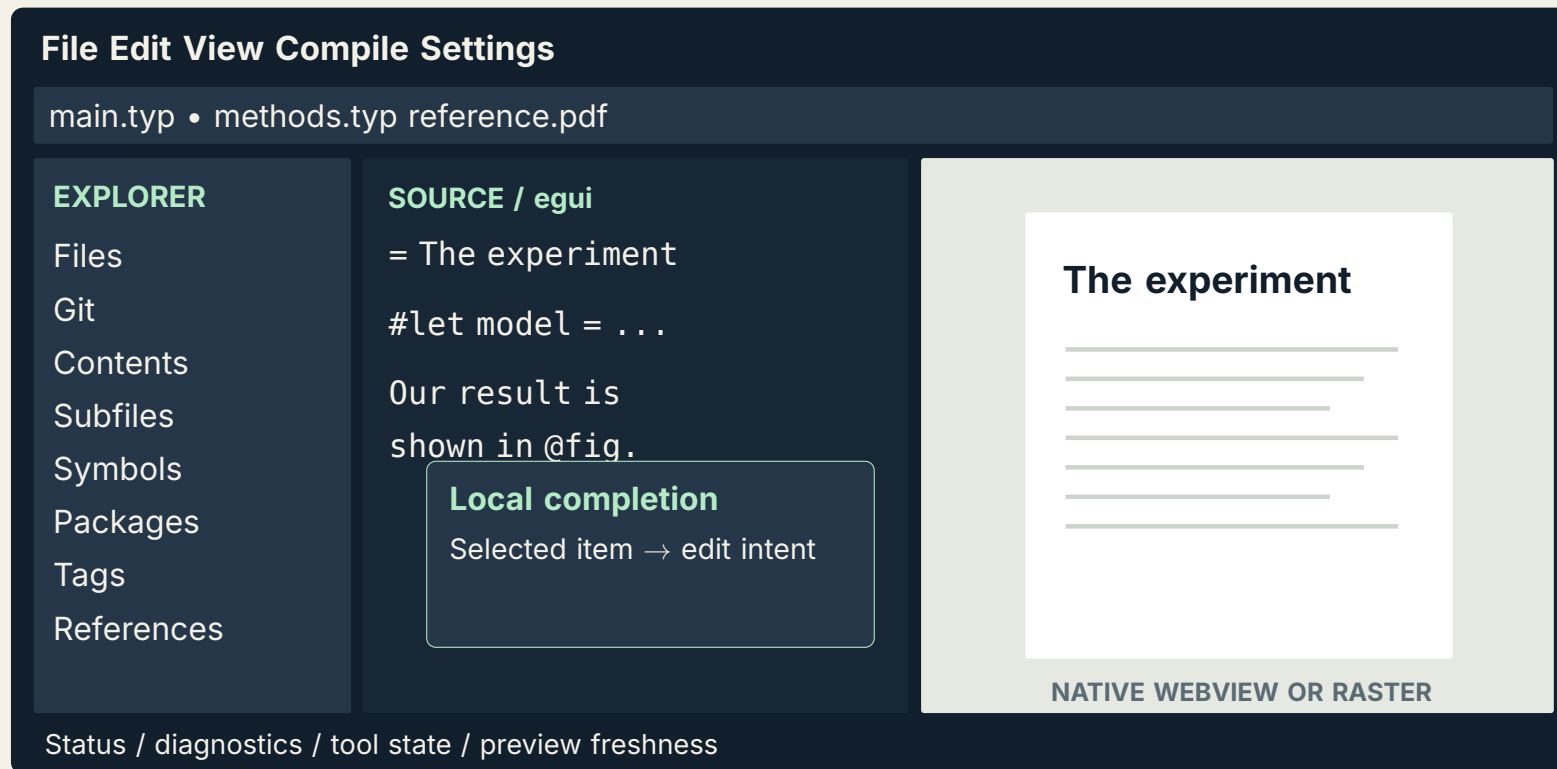
These counts describe different registries. Do not add them into a fictitious total-window count.

COUNT	WHAT IS COUNTED	SOURCE OF TRUTH
3 / 4	View modes / document kinds	Code, Split, Preview / Typst, Text, PDF, Image
4	Auxiliary tool-window families	Settings, Keyboard shortcuts, Packages, Typst overrides
8 / 5	AppPopup variants / AppModal variants	Application popup registry / document workflow modal enum
4 / 4	Child roles / child lifecycle states	ChildViewRole / ChildViewLifecycle
8 / 5 / 36	Explorer sections / settings sections / settings targets	ExplorerSection / SettingsSection / SettingsTarget
29 / 76 / 43	Menu commands / shortcut actions / QA scenes	AppCommand / ShortcutAction::ALL / UiSnapshot Scene::ALL

SOURCE src/app.rs · src/document.rs · src/app/settings_view.rs · src/child_view.rs · src/explorer.rs · src/native_menu.rs · src/shortcuts.rs · src/screenshot.rs

A workbench, not one undifferentiated canvas

Schematic only: this is a contributor map, not an application screenshot.



CHROME

Menus, toolbar, tabs and status are not extra OS windows.

PANELS

Explorer, Problems and Find are embedded surfaces.

OUTSIDE THIS FRAME

Tool windows, native pop-ups, modal overlays and OS dialogs have separate lifetimes.

Native child viewports can sit above the embedded preview; local egui overlays cannot be assumed to do so.

SOURCE `src/app/editor_view.rs · src/app/explorer_view.rs · src/app/raster_view.rs · src/app/native_views.rs`

Document windows and auxiliary tool windows

There can be many instances. Family count is not instance count.

FAMILY	OWNERSHIP / BEHAVIOR	FIRST FILE TO OPEN
Document window	Root or secondary EditorApp, coordinated by AppShell.	src/windowing.rs
Settings	Application-wide settings; primary host coordinates changes.	src/app/settings_window.rs
Keyboard shortcuts	Independent persistent viewport for search/chord capture.	src/app/settings_view.rs
Packages	Persistent package-catalog browser; emits requested actions.	src/app/native_views.rs
Typst overrides	Independent syntax-style editor; direct viewport path.	src/app/settings_view.rs
Native file chooser	OS-owned async dialog, not an application-rendered tool window.	src/workflow.rs / rfd adapter call sites

SOURCE [src/windowing.rs](#) · [src/app/settings_window.rs](#) · [src/app/settings_view.rs](#) · [src/app/native_views.rs](#) · [src/workflow.rs](#)

All eight AppPopup variants

One popup family has several content modes; completion and tooltips are not members of this enum.

VARIANT	WHAT THE USER IS DOING	WHERE CONTENT / POLICY LIVES
File / Edit / View	Invoking familiar application commands.	native_menu.rs metadata + app/native_views.rs
Workspace	Acting on a filesystem item or workspace context.	app/workspace_view.rs + app/native_views.rs
Editor	Acting on source selection/caret context.	app/editor_view.rs + app/native_views.rs
FontSelector	Choosing a document font through source-aware actions.	font_catalog.rs + app/native_views.rs
GitChunk	Inspecting a diff chunk and invoking a hunk action.	git/editor.rs + app/git_actions.rs
StatusLog	Inspecting accumulated application status.	app/native_views.rs + app.rs state

SOURCE src/app.rs · src/app/native_views.rs · src/native_menu.rs · src/app/git_actions.rs

All five workflow-modal variants

Rename and table editing are additional modal surfaces, not sixth/seventh AppModal variants.

VARIANT	DECISION	STATE THAT MUST SURVIVE
DeleteFile	Confirm removal of a chosen file.	Validated target path and workspace ownership.
UninstallPackage	Confirm deletion of a local package installation.	Specific installation identity, not only package name.
Alert	Acknowledge a notice or failure.	Title, message and notice severity.
Unsaved	Save, discard or cancel a deferred document action.	DocumentKey and the continuation's intended action.
Overwrite	Authorize replacing externally changed content.	Expected/observed disk fingerprints and document identity.

SOURCE src/workflow.rs · core/src/workflow.rs · src/app/native_views.rs

Transient surfaces: a practical inventory

These are functional families, not a single enum or a claim about simultaneous window count.

SURFACE	HOSTING CHARACTER	COMMON FAILURE MODE
Application menus / chunks / status	Native popup overlay; dismiss on blur.	Old popup action applied after its source changed.
Workflow / rename / table editor	Modal-style overlays; preserve pending decisions.	Treating focus loss as implicit cancellation.
Workspace chooser	Dedicated workspace-selection overlay.	Wrong root, stale MRU or incorrect native anchoring.
Diagnostic / function hover	Interactive tooltip child viewport.	Pointer bridge collapses; scrolling dismisses the card.
Asset hover / logo-color	Separate transient preview/color surfaces.	Stale texture or owner-unscoped color state.
Completion / settings hints / pickers	Local egui Areas/widgets in their host.	Assuming a local layer can overlay the native web-view.

SOURCE src/app/native_views.rs · src/app/tooltips.rs · src/window_logo.rs · src/app/completion_popup.rs · src/app/settings_controls.rs

Four child roles; four focus policies

Use the existing vocabulary when adding a surface.

ROLE	DEFAULT INTENT	FOCUS POLICY VOCABULARY
Persistent	A tool window can remain open and repaint independently.	Preserve
Modal	A pending decision blocks an owner workflow.	SuspendOnBlur
DismissOnBlur	A transient menu ends when interaction leaves it.	DismissOnBlur
Tooltip	A hover card can hand off into direct interaction.	InteractiveHandoff

Lifecycle is not a visibility boolean

The registry distinguishes four states: Visible, TemporarilyHidden, DurablyClosed and DormantHosted.

VISIBLE

The surface is presented and can participate in input. Its native identity, capture target and owner are all part of its contract.

TEMPORARILY HIDDEN

A modal or tooltip-related surface may suspend during focus transitions. Pending state remains meaningful; hiding is not permission to destroy it.

DURABLY CLOSED

The user or owner has ended the surface's lifetime. Late callbacks must not recreate it or apply actions to a replacement generation.

DORMANT HOSTED

A retained hidden host can support application lifecycle/reopen behavior. In particular, no visible document window does not necessarily mean the process has quit.

SOURCE `src/child_view.rs · src/windowing.rs · src/app/settings_window.rs`

Eight Explorer sections, exactly

Order is persisted by stable section identity, not by the last painted row position.

SECTION	CONTENT	CHANGE ROUTE
Files / Git	Filesystem tree / repository workflow.	workspace_service.rs + git.rs
Contents	Heading outline.	project_index.rs → app/explorer_view.rs
Subfiles	Literal local import/include dependencies.	project_index.rs path traversal
Symbols	Definitions and functions.	project_index.rs AST visitor
Packages	Imported package identifiers.	project_index.rs; browser is separate
Tags / References	Label definitions / reference occurrences.	project_index.rs → navigation.rs

SOURCE `src/explorer.rs · src/project_index.rs · src/app/explorer_view.rs · src/app/navigation.rs`

Five settings sections; thirty-six jump targets

A target is a searchable destination, not necessarily a separate screen.

SECTION / GROUP	TARGETS	IMPLEMENTATION ROUTE
Appearance	Appearance, TypstSyntax, LightTheme, DarkTheme, Invert Colors, HueShift, ThemeColors, PageTheme.	settings_panel.rs + presentation.rs
Editor behavior	WrapLines, LineNumbers, StickyContextRows, AutoPair Delimiters, MitexDollars, GitDiffStyle, RainbowBrackets, Auto Save, AutoSaveDelay.	settings.rs + editor adapters
Editor / interface	KeyboardShortcuts, ExplorerOrder, InterfaceScale, TitleBar Menus, FixedTabWidth, UiFont, UiFontWeight, CodeFont, Code FontWeight.	settings_controls.rs + theme.rs
Interaction / tools	PreviewJump, HoverDelay, TypstCompiler, TinymistLanguage Server, RefreshBinaryStatus, BrowseTypstPackages.	settings_view.rs + toolchain.rs
Preview / status	PreviewBackend, ToolchainStatus, ProjectRoot, UiScreenshots.	capabilities.rs + screenshot.rs

SOURCE [src/app.rs](#) · [src/app/settings_panel.rs](#) · [src/app/settings_controls.rs](#) · [src/settings.rs](#)

Commands, shortcuts and scenes are different axes

29 menu commands • 76 shortcut actions • 43 registered screenshot scenes.

COMMAND REGISTRY

AppCommand metadata defines menus, sections and availability requirements. Native menus and in-app menu content should consume the same semantics.

SHORTCUT REGISTRY

ShortcutAction also covers compile controls, folds, zoom, page navigation, tab selection and captures. Not every action is a menu item or has a default chord.

CAPTURE REGISTRY

UiSnapshotScene describes deterministic states to capture. Several scenes share one viewport; one viewport can have many themes and fixture states.

COVERAGE IS NOT A COUNT TRICK

43 scenes do not mean 43 windows or complete coverage. The gallery documentation describes 68 images from a selected theme/scene matrix, not every possible combination.

SOURCE [src/native_menu.rs](#) · [src/shortcuts.rs](#) · [src/screenshot.rs](#) · [docs/ui-qa-screenshots.md](#)

The architecture from orbit

One native shell; window-scoped services; a pure core where it counts.

- Repository and runtime boundaries
- Who owns what
- Two pipelines and the edit-to-output journey

Repository boundaries: policy, adapters, delivery

The root package has a binary and reusable library; core is a workspace member; xtask is separate.

ROOT PACKAGE / native application

src/main.rs + src/app.rs + app/* + windowing + services: eframe, native effects, workers, files, subprocesses.



ROOT LIBRARY / reusable transformations

src/lib.rs: miTeX document/projection, save transactions and theme-related exports.



CORE / pure policy

document · text · pairing / connection · preview · recovery / closing · workflow · geometry · scheduling

XTASK / separate

Tool downloads, packaging and repeatable QA/performance workflows.

EXTERNAL BOUNDARIES

Typst · Tinymist · Poppler /
Filesystem · Git · OS windows /
Networked package metadata

SOURCE Cargo.toml · src/main.rs · src/lib.rs · core/src/lib.rs · xtask/Cargo.toml

The source tree by responsibility

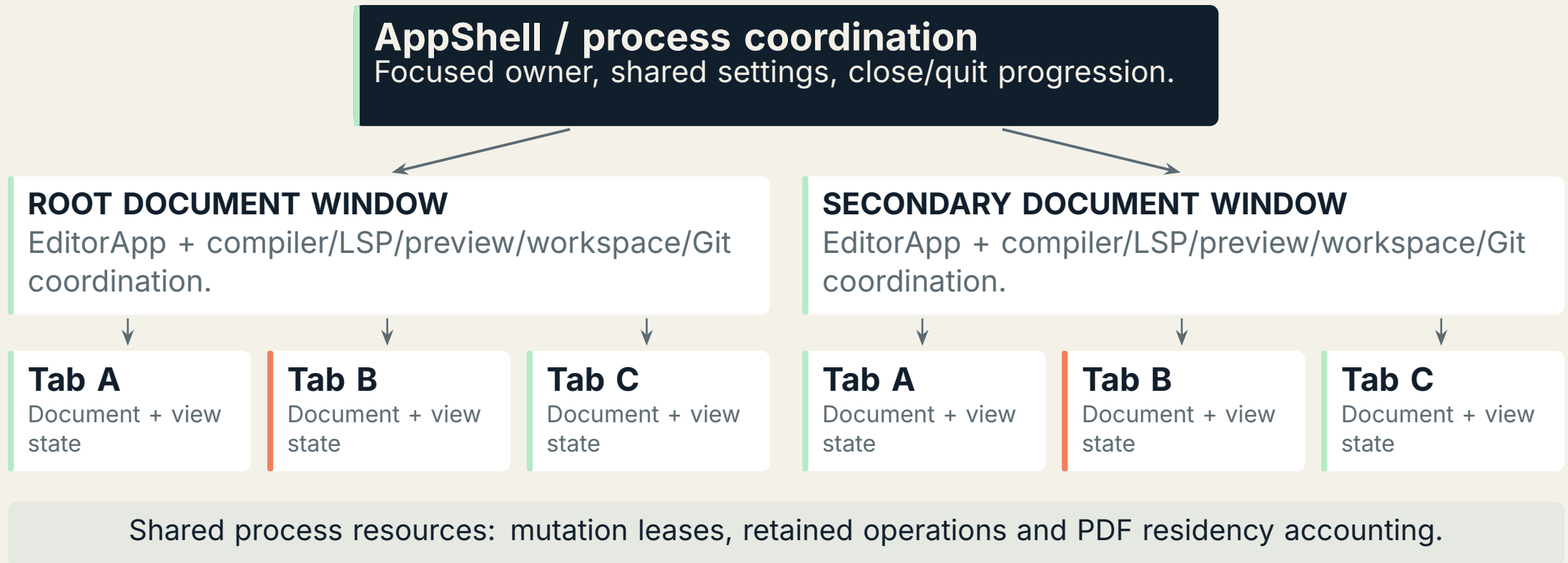
Start with the seam your change crosses, not the largest file.

AREA	ROLE	REPRESENTATIVE FILES
Boot / shell	Process startup, native events and window co-ordination.	main.rs, launch.rs, windowing.rs
Application adapters	Connect document state, workers and rendered surfaces.	app.rs and app/*.rs
Pure core	Versioning, geometry, recovery, scheduling, workflow policy.	core/src/*.rs
Source intelligence	Parsing, derived data, completion and projection.	editor_data.rs, project_index.rs, mitex_*.rs
External services	Subprocesses, filesystem and artifact producers.	compiler.rs, tinymist.rs, git/, workspace_service.rs
Delivery / evidence	Pinned tools, packaging, captures, performance fixtures.	toolchain/, xtask/, docs/, screenshot.rs

SOURCE src/main.rs · core/src/lib.rs · Cargo.toml · AGENTS.md

Ownership: one process is not one document

AppShell → document-window sessions → tab records. Services live at the window/session boundary.



SOURCE `src/windowing.rs · src/windowing/document_host.rs · src/app/tabs.rs · src/tinymist_sync.rs`

Shared state versus window-local state

Two windows can edit the same path without being the same document.

APPLICATION-WIDE

AppShell coordinates native commands, close/quit progression and shared settings. Global resource protection and PDF residency accounting serve multiple windows.

TAB-LOCAL

Stable tab IDs point to document records and editing/view state. Tab order, active tab and designated preview tab are separate pieces of state.

WINDOW-LOCAL

An EditorApp owns a document session/workspace context, preview/compiler/language-service coordination and Git/UI state. A secondary viewport has its own owner identity.

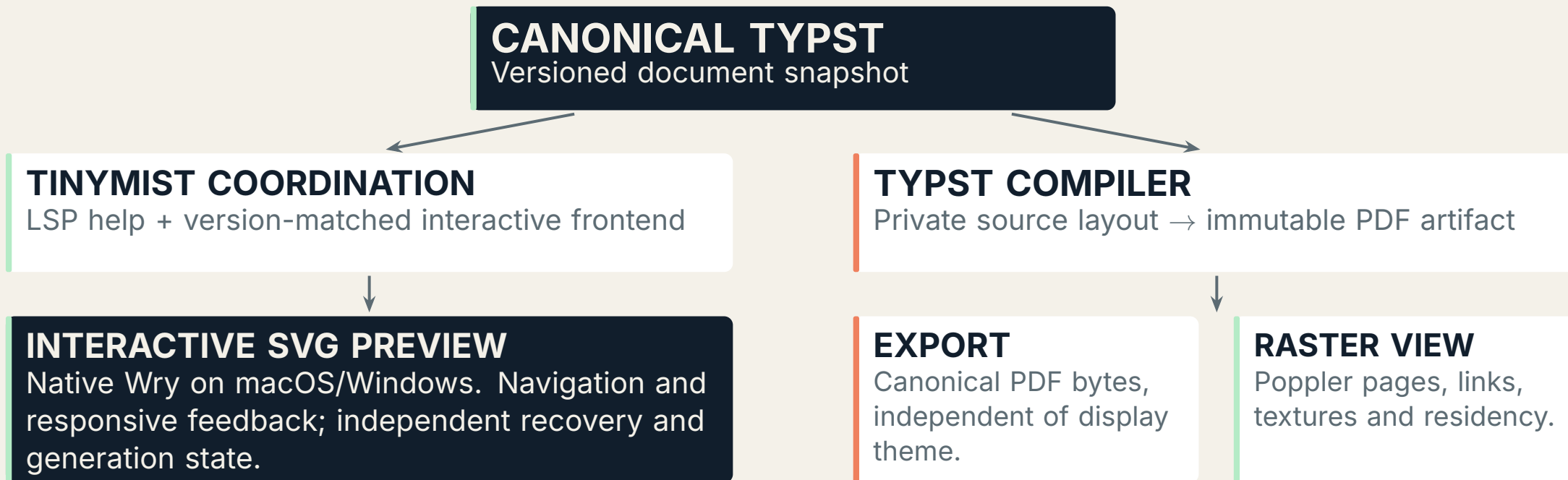
WHY THIS MATTERS

The same path and revision number can occur in two owners. A late result must never be accepted solely because its filename happens to match the visible tab.

SOURCE `src/windowing.rs · src/app/tabs.rs · src/resource_lock.rs · src/pdf_residency.rs`

Two preview routes, one canonical document

Keep rendering convenience separate from the artifact a user exports.



PAUSED PREVIEW $\neq$ DISABLED EDITING OR LSP • DISPLAY TRANSFORM $\neq$ DOCUMENT MUTATION

Interactive frontend transport and canonical PDF production have independent failure and freshness states.

SOURCE `src/compiler.rs · src/tinymist.rs · src/tinymist_sync.rs · src/pdf_pages.rs`

One edit fans out into several different jobs

The UI thread owns admission and acceptance; workers own blocking work.

DOCUMENT CHANGE

Capture an edit through the versioned document model. Preserve undo/selection intent; derive canonical text through the projection adapter when necessary.

LOCAL DERIVATION

Invalidate source-dependent parsing, highlighting, search, folds and structural data. Repaint-only changes should not rebuild source-only indexes.

SERVICE COORDINATION

Schedule compiler and index work, synchronize Tinymist document changes and refresh buffer-aware Git information. Superseded results are not applied blindly.

USER-VISIBLE OUTCOME

Accepted results update preview/diagnostics/navigation. Saving is a separate transaction; a live preview edit is not automatically a write to the user's file.

SOURCE `core/src/document.rs · src/editor_data.rs · src/index_jobs.rs · src/tinymist_sync.rs`

The app module is an adapter, but still substantial

Read extracted modules before deciding where the next extraction belongs.

EXTRACTED UI

Editor, Explorer, settings, completion, find bar, tooltips, packages, tabs and raster rendering already have dedicated app submodules.

EXTRACTED POLICY

Document workflow, core state machines, synchronization, save transactions and worker abstractions already exist. Do not rebuild a parallel architecture to solve one feature.

REMAINING GRAVITY

app.rs still joins many concerns and includes extensive inline tests. Its large file size is a maintainability signal—not a measurement of production complexity alone.

GOOD EXTRACTIONS

Move one cohesive state machine or borrowed-input/typed-output renderer. Preserve behavior and tests first; change the product in a subsequent patch.

SOURCE `src/app.rs · src/app/explorer_view.rs · src/app/settings_window.rs · src/app/completion_popup.rs`

Startup and native lifecycle are a separate subsystem

Platform bridges are small in concept and easy to destabilize.

BOOT SEQUENCE

Launch options and build/version handling precede AppShell creation. Native menu and file-open requests enter queues rather than mutating arbitrary editor state directly.

ROOT WINDOW

The native shell configures the root viewport and restores suitable window dimensions. A retained host participates in macOS close/reopen semantics.

MACOS BRIDGE

The integration augments the event-loop delegate rather than replacing its lifecycle. The manifest pins the winit release used by that bridge.

CONTRIBUTOR RULE

Treat unsafe/native code as a narrow boundary. Verify native events, focused-owner dispatch, reopen and quit separately from headless widget behavior.

SOURCE `src/main.rs` · `src/open_requests.rs` · `src/native_menu.rs` · `Cargo.toml`

Dependencies: what each major piece buys you

Manifest snapshot, not a recommendation to upgrade every crate at once.

DEPENDENCY	PURPOSE	UPGRADE RISK TO CHECK
eframe / egui + glow	Immediate-mode desktop UI, persistence, viewports.	Input/focus, fonts, deferred viewport delivery and captures.
Wry	Embedded interactive frontend on macOS/Windows.	Native bounds, visibility, lifecycle and webview ownership.
typst-syntax / syntect	Typst AST and general-language highlighting.	Syntax roles, parsing, incremental data and theme semantics.
notify / rfd	Filesystem notifications / native async dialogs.	Watcher gaps, root changes and late dialog results.
serde / JSON / TOML / XML	Persistence and protocol/configuration parsing.	Untrusted/partial data and schema evolution.
skrifa / image / URL / HTTP	Fonts, assets and package/link handling.	Resource limits, failure reporting and capability separation.

SOURCE Cargo.toml · src/toolchain.rs · src/capabilities.rs

The contracts below the UI

The details that prevent stale results, broken edits and lost work.

- Identity and text coordinates
- Compile, preview and background lifetimes
- Saving, Git, indexing and native surfaces

Identity: the counters are not interchangeable

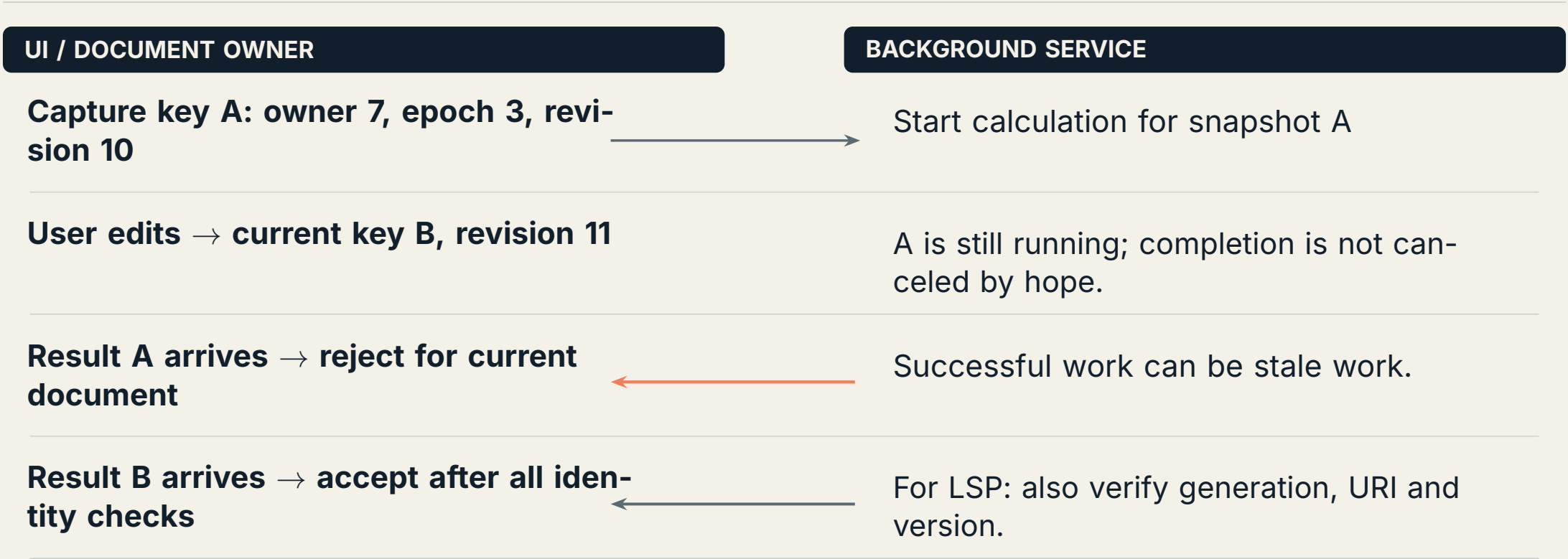
Use the identity belonging to the operation—not whichever counter is easiest to reach.

IDENTITY	WHAT IT DISTINGUISHES	FAILURE IT PREVENTS
WindowSessionId	Two document-window owners in one process.	A result for window A mutates window B.
DocumentKey: owner + epoch + revision	Owner; document replacement/reopening; source edit.	Same path and revision mistaken for the same document.
TabId	Stable record identity, independent of visual tab order.	A reorder changes the meaning of an index.
LSP generation + URI + version + key	A particular server incarnation and synchronized document.	A restarted server or closed/reopened URI supplies stale help.
ArtifactKey	Source revision and compiler generation for an artifact.	An old successful render replaces newer output.
Child-view generation	A particular opening/lifetime of a native surface.	A deferred callback closes or acts on its replacement.

SOURCE core/src/document.rs · core/src/preview.rs · src/app/tabs.rs · src/tinymist_sync.rs · src/child_view.rs

The stale-result rule, made concrete

A worker can finish successfully and still have no right to update the current UI.



SOURCE core/src/document.rs · src/tinymist_sync.rs · src/windowing/document_host.rs

Text has more than one coordinate system

Illustrative string: a + U+1F600 grinning face + é. Offsets below are insertion boundaries.

POSITION	UTF-8 BYTES / UNICODE SCALARS	UTF-16 COLUMN
Before a	0 bytes / 0 scalars	0
After a	1 byte / 1 scalar	1
After U+1F600	5 bytes / 2 scalars	3
After é	7 bytes / 3 scalars	4
Editor ↔ Rust strings	egui cursor positions are scalar-based; Rust slicing is byte-based.	Use the typed conversion layer.
LSP ↔ preview navigation	LSP uses UTF-16; preview line/column navigation is scalar-based.	Do not reuse one conversion for both protocols.

SOURCE `core/src/text.rs`

Safe edits are a transaction, not a loop

A server edit batch has to be valid as a whole before any source changes.

PREPARE

Choose the coordinate space and capture DocumentKey. Build one source index for the batch; convert every range against the same immutable input.

VALIDATE

Reject reversed, overlapping or invalid ranges, including a UTF-16 boundary inside a surrogate pair. Navigation can clamp; a destructive edit must be stricter.

PROJECT

For miTeX display mode, respect display-versus-canonical coordinates. A legal edit in one representation is not automatically legal in the other.

COMMIT

Recheck the key, apply the prepared result once and preserve the intended cursor/undo operation. A stale transaction must leave the current document untouched.

Derived data: make unchanged frames cheap

EditorDerivedData is a revision-keyed cache, not a bag of incidental memoization.

SOURCE-KEYED

Character starts, source metrics and the immutable source handle rebuild only when the document key changes. Syntax preparation is lazy.

QUERY-KEYED

Delimiter and hover queries include the caret position. Structural regions are shared by folding and sticky context rather than independently reparsed for each consumer.

DIAGNOSTIC-KEYED

Diagnostic preparation also includes generation, current/virtual paths and whether this source drives preview. Similar text is not sufficient identity.

TEST THE WORK

Existing tests count rebuilds and compatibility checks. For a cache change, assert which work is skipped; avoid relying only on a noisy elapsed-time claim.

Highlighting, folding and sticky context

These are related views over source, but their responsibilities differ.

HIGHLIGHTING

Typst syntax and generic text highlighting have separate adapters. Theme role resolution should be shared; document-specific semantics should not leak into unrelated languages.

FOLDING

Hidden rows have zero-height layout/no painted mesh while source coordinates remain intact. Folding does not remove text from the saved buffer.

CURSOR BEHAVIOR

Vertical movement can skip concealed lines; navigation to a hidden location must reveal the target rather than silently send the user somewhere else.

CONTEXT

Headings, bindings, functions and blocks provide structural ranges. Preserve fold state across compatible edits; do not associate it with a completely different document epoch.

SOURCE [src/highlight.rs](#) · [src/generic_highlight.rs](#) · [src/folding.rs](#) · [src/editor_features.rs](#)

Completion: from candidate list to one undo step

Ranking, rendering and source mutation are deliberately different stages.

RANK

Case-insensitive subsequence matching rewards early, consecutive matches. Filter before the 200-item display limit so good late candidates are not discarded prematurely.

ADAPT

Reference labels, font-string contexts and TeX candidates need local interpretation. Rebase only when a compatible prefix edit leaves the rest of the source unchanged.

RENDER

`completion_popup.rs` receives borrowed display data and returns selection/accept/dismiss actions. It is a local egui Area, not a native tool window.

APPLY

`completion_edit.rs` prepares main and additional edits, expands snippet text and validates identity before committing. Do not mutate the document from the row renderer.

SOURCE `src/completion.rs · src/tex_completion.rs · src/app/completion_popup.rs · src/completion_edit.rs`

miTeX projection: an adapter with refusal modes

The display can be convenient only while its mapping remains trustworthy.

CANONICAL MODEL

The source document stays valid Typst. Dollar-form TeX is an editable display projection over supported miTeX calls, not a second independently saved file format.

BINDING AWARENESS

The projection checks the configured miTeX package/import and renderer binding. Shadowed or conflicting bindings cannot safely be treated as the intended renderer.

BOUNDARY MAPPING

Mappings must preserve order and valid UTF-8 boundaries. Ambiguous native math, malformed source or an unclosed projected expression must not be guessed into a rewrite.

TEST SURFACE

Exercise source→display→source, insertion at every boundary, Unicode, undo/redo, mode switching, imports and completion edits. Refusal without mutation is a successful safety outcome.

SOURCE `src/mitex_projection.rs · src/mitex_document.rs · src/app/mitex_mode.rs`

Table editing is intentionally conservative

Static source transformation—not a miniature arbitrary Typst evaluator.

RECOGNIZE

The structural helper recognizes supported table/grid calls and rectangular cell arrangements. It retains non-structural options and the content of cells.

REFUSE

Dynamic/spread arguments, computed columns, structural helpers, malformed syntax and ambiguous argument comments can make a safe round trip impossible.

EDIT A DRAFT

A native modal owns a draft table. Add/remove rows or columns without immediately writing into the source document; cancellation leaves source unchanged.

APPLY ONCE

Validate the originating document key/range before replacing the call as one edit. Test re-open, stale source, focus loss, cancellation and undo—not just cell rendering.

Tinymist synchronization: more than `didChange`

The coordinator tracks server lifetime and which documents must remain available.

PLAN EFFECTS

`tinymist_sync.rs` computes open/change/close and backing-file effects from immutable canonical snapshots. Keep protocol sequencing separate from collecting application state.

ACTIVE $\neq$ PREVIEW

A pinned preview entry can stay open while another tab is edited. Changing the active tab must not close the source that still drives preview.

ACCEPT REPLIES

Require generation, URI, version and `DocumentKey` to agree. URI/version alone cannot distinguish a document that was closed and opened again.

STOP IN ORDER

Send required close effects before releasing backing resources. Restart and shutdown are lifetime transitions, not merely a boolean that hides the preview.

SOURCE `src/tinymist_sync.rs` · `src/tinymist/transport.rs` · `src/tinymist/protocol.rs`

The compiler publishes immutable evidence

PDF generation must not be held hostage by page decoding or display state.

PRIVATE INPUT

Build against a private source layout under the chosen root context. Canonical source, relative dependencies and fonts belong to the request—not to whatever tab is visible later.

WATCHER LIFETIME

The worker supports a persistent watch context; macOS restarts for buffer updates to avoid the documented dot-directory notification problem. Stale sessions/logs are rejected.

ARTIFACT FIRST

Read a produced PDF into an immutable byte snapshot and publish it before optional catalog/raster work. Export-only work need not require Poppler.

CLEAN SHUTDOWN

Stop/reap subprocesses, bound reader shutdown and remove private material at the right lifetime boundary. Killing a UI receiver is not a substitute for process cleanup.

SOURCE `src/compiler.rs · src/private_workspace.rs · src/process.rs`

PDF pages: freshness and residency are separate

An accepted document artifact can still have pages that are not decoded or uploaded yet.

CATALOG / DECODE / DRAW

PDF metadata and page links, page raster jobs and texture presentation have distinct work. A new artifact invalidates the old artifact's page identity.

BOUNDED RESIDENCY

The inspected accounting has process-wide budgets of 96 MiB decoded data and 192 MiB textures. Multiple windows participate in the same accounting.

NOT A TOTAL MEMORY CAP

Those numbers are not a bound on process RSS, browser memory, fonts or all GPU allocations. Measure the resource whose budget is being discussed.

DISPLAY-ONLY TRANSFORMS

Dark/inverted page rendering changes the displayed image, not the compiler's PDF bytes. Test artifact export independently from transformed preview pixels.

SOURCE `src/pdf.rs · src/pdf_pages.rs · src/pdf_residency.rs · src/app/raster_view.rs · src/preview.rs`

Recovery should be boring and bounded

Crashing preview transport must not turn into an infinite restart storm.

PURE POLICY

The core recovery model takes an explicit generation and injected time. This makes retry decisions deterministic and testable without sleeping.

BUDGET

The inspected policy has a five-failure limit and one-second retry delay: four retries, then fallback. Old-generation and duplicate failure notifications do not consume arbitrary extra budget.

CONSUME ONCE

A scheduled retry can be taken once. A genuine accepted success resets the failure budget; a merely arriving but invalid ready message must not.

KEEP OTHER CAPABILITIES

Fallback does not mean that editing, manual PDF generation or language services should disappear. Render and report each capability's actual availability.

Background work: choose the correct lifetime

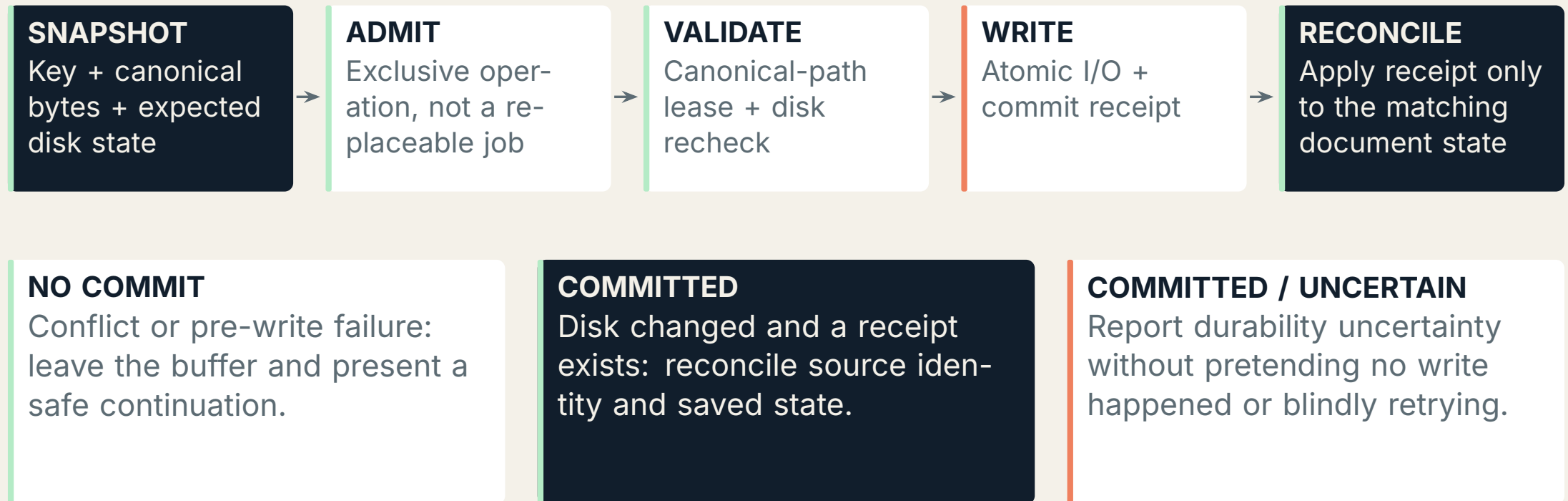
“Async” is not a safety policy.

PATTERN	SUITABLE WORK	CONTRIBUTOR OBLIGATION
LatestJob	Replaceable calculations whose old result is no longer useful.	Dropping an old receiver does not necessarily interrupt running code.
Latest-job queue	A worker that should take the newest queued request.	Bound backlog; validate result identity after processing.
Exclusive operation	Save, Git or other externally visible mutations.	Do not supersede a write; reject duplicate admission while active.
RepaintTarget	Wake the owning viewport after background work.	Capture the intended owner early; never mutate UI state from a worker.
Service polling	Drain valid events and detect disconnection.	Report a terminal disconnect once, after pending events are handled.
Detached completion	Mutation finishing after its initiating window closed.	Retain outcome/operation lifetime; let close/quit coordination observe it.

SOURCE [src/worker.rs](#) · [src/worker/latest_queue.rs](#) · [src/worker/exclusive.rs](#)

Saving: ownership survives the write

A proposed mental model of the implemented save transaction and I/O boundaries.



SOURCE `src/save_transaction.rs · src/save_io.rs · src/resource_lock.rs · src/app/saves.rs`

A save result can be committed but uncertain

Do not collapse all non-happy outcomes into "try again".

INPUT SNAPSHOT

SaveInput owns the immutable request, expected disk state, intent and continuation. Explicit, confirmed-overwrite and automatic saves have different admission rules.

PRECOMMIT CONFLICT

Recheck the disk fingerprint while holding the application's canonical-path resource lease. Missing and unchecked disk states are distinct cases.

COMMIT RECEIPT

A successful write produces a receipt. Synchronized versus uncertain durability is meaningful: a post-commit durability concern is not an ordinary pre-write failure.

EXTERNAL RACES

The resource lease serializes this process's operations. It is not an OS-wide lock and cannot prevent an unrelated editor changing the same file at every possible moment.

Closing and quitting are workflows, not destroy()

A confirmation applies to a particular document state.

DIRTY DOCUMENTS

Unsaved prompts carry document identity and a pending action. A newly edited document can invalidate a previously obtained close permit.

ACTIVE MUTATIONS

Closing a window must not orphan an admitted save/Git operation. Process-level operation tracking and retained completions allow the shell to wait or report outcomes.

NATIVE DIALOGS

Open/save dialogs can return late. Revalidate their owner and document context before applying a path or continuation to the currently visible editor.

MACOS NO-WINDOW STATE

Closing the root document can leave a dormant clean host rather than quitting the app. Dock reopen, settings and real Quit have different semantics.

SOURCE [src/workflow.rs](#) · [core/src/closing.rs](#) · [src/windowing.rs](#) · [src/worker/exclusive.rs](#)

Git has three useful baselines

Saving, staging and committing are not interchangeable.

EDITOR GUTTER

Buffer-aware changes can include unsaved text. Keep editor hunk identity tied to the source snapshot, not just the file on disk.

REPOSITORY VIEWS

Unstaged changes compare working tree with index; staged changes compare index with HEAD. Different views can legitimately show different hunks.

STAGE / UNSTAGE

Index mutations run under the repository transaction boundary. Validate paths, current state and conflicts; refuse unsafe overlapping staged hunk transformations.

REVERT IN EDITOR

A revert verifies the expected current source and returns a new buffer value. The editor applies an undoable edit; that action is not an implicit write to the working file.

SOURCE `src/git/editor.rs · src/git/repository.rs · src/git/repository/hunks.rs · src/app/git_actions.rs`

Project indexing is bounded structural analysis

Fast local navigation does not imply full semantic evaluation.

INPUTS

Parse source structure, follow supported literal local imports/includes and incorporate relevant unsaved source overrides. Track visited paths to stop cycles.

BOUNDARIES

Canonicalize and enforce the project root. Do not execute arbitrary Typst or download packages to discover a sidebar dependency.

COMPLETENESS

The inspected cap is 256 project files. File-limit and unreadable-file conditions are represented as partial-index warnings; dynamic dependency expressions can remain unresolved.

CANCELLATION

A root/source change can supersede indexing. Preserve useful existing results while new work runs, but never display an old root's locations as the new root's index.

SOURCE `src/project_index.rs · src/index_jobs.rs · src/workspace_service.rs`

Capabilities and tool resolution belong off the paint path

A missing optional tool should produce a precise limitation, not a frozen frame.

SIX CAPABILITIES

Editing, language server, interactive preview, PDF generation, rasterization and link extraction are assessed separately.

POPPLER IS SPLIT

Rasterization needs pdfinfo and pdftoppm; pdftohtml supplies link extraction. Missing links and inability to rasterize are different failures.

TOOLCHAIN

Pinned Typst/Tinymist sidecars and user/environment/system discovery are managed in toolchain.rs. Invalid overrides should remain visible in status even when a fallback resolves.

REFRESH EXPLICITLY

Cache tool/capability discovery; refresh when configuration or an explicit action changes it. Never search PATH or launch a version probe for every painted frame.

SOURCE [src/capabilities.rs](#) · [src/toolchain.rs](#) · [docs/architecture/0003-bundled-toolchain.md](#)

Native overlays: focus is geometry plus time

The tricky bugs often happen between two windows, not inside either widget.

DECLARED CONTRACT

A child surface declares role, focus behavior, bounds and lifetime. The owner accepts only actions from the correct live generation.

TOOLTIP HANDOFF

A native hover needs a stable corridor from the source token to the popup. The safe-triangle apex remains at the original pointer; a grace interval spans the native gap.

INTERACTION

Some tooltips permit scrolling and text selection after handoff. A modal can suspend on blur rather than dismissing an unfinished edit.

LOCK DISCIPLINE

Derive viewport identity before taking egui Context data locks. A convenience query inside the lock can re-enter context access and deadlock.

SOURCE `src/child_view.rs · src/app/tooltips.rs · src/app/popup_layout.rs`

Themes and fonts cross every window boundary

The visual system is a model; each surface is a consumer.

SEMANTIC STYLE

theme.rs centralizes colors, spacing, radii and reusable controls. Syntax themes and override roles add source-specific presentation, not arbitrary widget-local constants.

PREPARED PRESENTATION

presentation.rs provides the resolved appearance consumed by application surfaces. Light/dark choices, transforms and typography must propagate consistently.

FONT SERVICES

Cataloging and previews belong in dedicated font modules. Unicode fallback and per-viewport font setup must cover detached surfaces as well as the main editor.

CACHE INVALIDATION

An appearance/font change should invalidate affected layout/paint caches without rebuilding source-only analysis. Conversely, stale cached geometry must not clip a newly larger tooltip.

SOURCE [src/theme.rs](#) · [src/presentation.rs](#) · [src/syntax_theme.rs](#) · [src/font_catalog.rs](#) · [src/viewport_fonts.rs](#) · [src/unicode_fonts.rs](#)

Tool selection: the exact precedence

A development environment variable does not outrank a valid selected or bundled tool.

ORDER	CANDIDATE	WHAT THE USER SHOULD SEE
1 / explicit Custom mode	A valid executable at the configured custom path.	Custom origin; a broken custom choice remains a visible problem.
2 / bundled candi- dates	Executable-adjacent, resource-style and development bundle locations.	Bundled origin, pinned version and any custom-path fallback reason.
3 / environment fall- back	TIPTOPTYP_TYPST or TIPTOPTYP_TINYMIST.	The variable is named in the fallback explanation.
4 / PATH fallback	Search the host executable path only after earlier candidates fail.	PATH origin; never silently pretend this is the pinned bundle.
5 / unavailable	No valid candidate exists.	Missing capability and an actionable resolution failure.

SOURCE src/toolchain.rs · src/settings.rs · toolchain/manifest.tsv

Operational constants worth knowing

Implementation defaults in the inspected source—not performance guarantees or targets for every machine.

POLICY	INSPECTED VALUE	WHY IT MATTERS
Work admission	Compile debounce 60 ms; project-index debounce 180 ms.	An edit can schedule different pipelines on different cadences.
Autosave retry / status history	Retry delay 2 seconds; history retains up to 100 entries.	Do not create a busy-loop retry or treat status history as durable logging.
Tinymist timeouts	Initialize 15 s; preview 30 s; interactive requests 10 s; formatting 120 s.	Different operations have different expected lifetimes.
Native hover handoff	300 ms grace interval.	A real pointer crosses a gap between native surfaces.
Bounded indexing / completion	256 project files; 200 displayed completion candidates.	Expose partial indexes and rank/filter before truncating candidates.
Decoded / texture residency	96 MiB decoded; 192 MiB textures, process-wide.	These counters do not bound total application or browser memory.

SOURCE [src/app.rs](#) · [src/tinymist.rs](#) · [src/app/tooltips.rs](#) · [src/project_index.rs](#) · [src/completion.rs](#) · [src/pdf_residency.rs](#)

"I want to change X. Where is Y?"

A source atlas organized by the effect a contributor wants to produce.

- Feature-to-file signposts
- The nearby contracts and tests
- Worked routes through a safe patch

Route map / source editing

Start at the smallest domain module; follow the adapter only when needed.

USER-FACING CHANGE	START HERE	THEN CHECK
Typing / selection / editor layout	src/app/editor_view.rs	core/src/text.rs; src/editor_data.rs
Auto-closing delimiters	src/auto_pairs.rs	core/src/pairing.rs; editor integration
Matching / rainbow brackets	src/delimiters.rs; src/rainbow.rs	src/highlight.rs; settings and themes
Sticky context / folds	src/editor_features.rs; src/folding.rs	editor_view.rs; navigation.rs
Find / replace behavior	src/search.rs; src/app/find_bar.rs	shortcuts.rs; Unicode/regex tests
miTeX dollar editing	src/mitex_projection.rs; mitex_document.rs	app/mitex_mode.rs; completion_edit.rs

SOURCE src/app/editor_view.rs · src/search.rs · src/folding.rs · src/mitex_projection.rs

Route map / language and structural tools

Separate "what source means" from "how the result is drawn".

USER-FACING CHANGE	START HERE	THEN CHECK
Completion content / ranking	src/completion.rs; tex_completion.rs	tinymist/protocol.rs; font_catalog.rs
Completion popup appearance	src/app/completion_popup.rs	theme.rs; popup_layout.rs; local Area tests
Applying completion / formatting edits	src/completion_edit.rs; core/src/text.rs	DocumentKey; miTeX mapping; one-step undo
Hover text / diagnostics	src/diagnostics.rs; app/tooltips.rs	tinymist.rs; editor_data.rs; owner/generation
Table/grid editor	src/editor_features.rs; app/native_views.rs	stale-source rejection; cancel/apply; source preservation
Asset or font inspection	src/asset.rs; font_preview.rs	editor_features.rs; tooltips.rs; native_views.rs

SOURCE [src/completion.rs](#) · [src/completion_edit.rs](#) · [src/app/tooltips.rs](#) · [src/editor_features.rs](#)

Route map / project and navigation

Explorer is the renderer; indexing and filesystem work live elsewhere.

USER-FACING CHANGE	START HERE	THEN CHECK
Explorer section order / identity	src/explorer.rs	settings.rs; app/explorer_view.rs
File tree row / context behavior	src/app/explorer_view.rs	app/workspace_view.rs; workflow.rs
Root switching / refresh / tree scan	src/workspace.rs; workspace_service.rs	app/lifecycle.rs; index_jobs.rs
Contents / symbols / tags / references	src/project_index.rs	src/index_jobs.rs; app/navigation.rs
Tabs / preview-entry choice	src/app/tabs.rs	tabs_tests.rs; tinymist_sync.rs
Source or preview navigation	src/app/navigation.rs	core/src/text.rs; preview.rs; tinymist.rs

SOURCE src/explorer.rs · src/app/explorer_view.rs · src/project_index.rs · src/app/tabs.rs

Route map / preview, PDF and tools

Ask whether you are changing the interactive view, the artifact or its raster presentation.

USER-FACING CHANGE	START HERE	THEN CHECK
Interactive preview behavior	src/preview.rs; tinymist.rs	tinymist_sync.rs; core/src/recovery.rs
Native webview geometry / visibility	src/app/native_views.rs	native_window.rs; child_view.rs; platform bridge
Compile / exported PDF bytes	src/compiler.rs	private_workspace.rs; app/saves.rs
PDF page rendering / links	src/pdf.rs; pdf_pages.rs	app/raster_view.rs; capabilities.rs
Page memory / cross-window residency	src/pdf_residency.rs	ArtifactKey; decoded-versus-texture counters
Tool paths / versions / availability	src/toolchain.rs; capabilities.rs	toolchain/manifest.tsv; xtask; Settings/Status

SOURCE src/preview.rs · src/compiler.rs · src/pdf_pages.rs · src/toolchain.rs

Route map / popups and panels

The word “popup” hides several implementations. Choose the right family.

SURFACE	START HERE	NEARBY CONTRACT
File / Edit / View / contextual menus	AppPopup in src/app.rs; app/native_views.rs	app/popup_layout.rs; child_view.rs
Status history / Git hunk / font selector	Their AppPopup branch + domain action producer	status history; git/editor.rs; font catalog
Diagnostic / function tooltip	src/app/tooltips.rs	hover timing, safe geometry, selection and capture
Settings hint / completion list	app/tooltips.rs / app/completion_popup.rs	Local egui Area; not a native child window
Problems / find / Git panel	app/editor_view.rs; find_bar.rs; git_actions.rs	diagnostics.rs; search.rs; git/
Unsaved / overwrite / delete / alert	src/workflow.rs; app/native_views.rs	DocumentKey; close/save continuation

SOURCE src/app.rs · src/app/native_views.rs · src/app/tooltips.rs · src/workflow.rs

Route map / independent windows and appearance

A detached viewport needs more than a new drawing function.

USER-FACING CHANGE	START HERE	THEN CHECK
Settings window	src/app/settings_window.rs; settings_panel.rs	settings_view.rs; shared settings merge
Keyboard shortcut editor	src/app/shortcut_editor.rs	shortcuts.rs; capture/conflict behavior
Package browser window	src/app/package_browser.rs	package_catalog.rs; native_views.rs
Typst syntax overrides	src/app/settings_view.rs; syntax_theme.rs	settings_controls.rs; presentation.rs
Colors / spacing / typography	src/theme.rs; presentation.rs	viewport_fonts.rs; unicode_fonts.rs
Window tint / native controls	src/window_logo.rs; native_window.rs	native_menu.rs; main.rs; platform tests

SOURCE src/app/settings_window.rs · src/app/shortcut_editor.rs · src/app/package_browser.rs · src/theme.rs

Route map / writes, Git and application lifetime

Anything that changes persistent state deserves its own failure story.

USER-FACING CHANGE	START HERE	THEN CHECK
Save / autosave / over-write conflict	src/app/saves.rs; save_transaction.rs	save_io.rs; resource_lock.rs; workflow.rs
Dirty tab/window close or Quit	src/workflow.rs; windowing.rs	core/closing.rs; worker/exclusive.rs
Git repository actions	src/git/repository.rs; app/git_actions.rs	repo resource lease; status refresh; error reporting
Editor hunk markers/ actions	src/git/editor.rs; git/repository/hunks.rs	buffer identity; undo; staged/unstaged boundaries
Native menu dispatch	src/native_menu.rs	shortcuts.rs; focused document owner in windowing.rs
Open requests / new windows	src/open_requests.rs; launch.rs	windowing/document_host.rs; stale dialogs

SOURCE src/app/saves.rs · src/save_transaction.rs · src/git/repository/hunks.rs · src/windowing.rs

Route map / debugging and delivery

Evidence should be reproducible by the next contributor.

PROBLEM OR CHANGE	START HERE	EXPECTED EVIDENCE
Geometry / clipping / focus	src/app/qa.rs; app/popup_layout.rs	Deterministic geometry + native lifecycle checks
Screenshot fixture / gallery	src/screenshot.rs; docs/ui-qa-screenshots.md	Correct viewport, fresh capture, known omissions
Sluggish interaction / allocation churn	src/performance.rs; docs/performance.md	Matched optimized run + deterministic work counters
Tool packaging / sidecar download	xtask/; toolchain/manifest.tsv	Hashes, supported triples, clear missing-tool errors
Build/version reporting	src/build_info.rs; Cargo.toml	Revision/dirty/timestamp metadata when built normally
Contributor process / architecture	AGENTS.md; docs/architecture/	Small cohesive patch, preserved behavior, updated docs

SOURCE src/screenshot.rs · docs/performance.md · src/build_info.rs · AGENTS.md

Worked patch / make a menu popup wider

A geometry change should not become an accidental lifetime rewrite.

1 / IDENTIFY THE FAMILY

Is the target AppPopup, a completion Area, a workflow modal or a persistent settings viewport? Reproduce one specific surface and record its owner, scale and screen bounds.

2 / CHANGE THE RIGHT LAYER

For native menu geometry, inspect popup_layout.rs and shared theme metrics before editing each menu branch. Keep content measurement and viewport placement in agreement.

3 / KEEP BEHAVIOR

Preserve dismiss-on-blur, click-outside, Escape, generation gating and initial focus grace. A larger popup still needs reachable content near every screen edge.

4 / PROVE THE PATCH

Add deterministic placement cases at mixed scales and edges; capture the matching scene; manually check native/Wry overlap and focus where framebuffer evidence cannot decide.

SOURCE [src/app/popup_layout.rs](#) · [src/child_view.rs](#) · [src/app/native_views.rs](#) · [docs/ui-qa-screenshots.md](#)

Worked patch / add a new Explorer index section

Do not put project traversal in the sidebar paint function.

1 / DEFINE THE MODEL

Add the stable section identity/order behavior in `explorer.rs`. Define the indexed item type and the source/root identity needed by navigation.

2 / PRODUCE DATA

Extend `project_index.rs` or a dedicated index job. Parse once, honor cancellation/root boundaries and expose completeness rather than silently skipping unsupported inputs.

3 / RENDER INTENT

Pass prepared rows into `explorer_view.rs`. Return a typed navigation/action result; let the owner select/open a tab and apply a converted position.

4 / CLOSE THE LOOP

Add settings-order handling, empty/loading/partial states and a deterministic fixture. Test unsaved input, Unicode locations, cycles, root switches and stale job completion.

SOURCE `src/explorer.rs · src/project_index.rs · src/app/explorer_view.rs · src/app/navigation.rs`

Worked patch / add a keyboard-accessible action

One action should not acquire three contradictory implementations.

1 / DECLARE

Choose a stable action ID and human label in shortcuts.rs. Decide availability and whether it also belongs in AppCommand/native menus.

2 / DISPATCH

Route keyboard, menu and any toolbar entry to the same owner-side action. Resolve the focused document window before executing a native menu event.

3 / PERSIST

Preserve explicit disabled bindings, sensible defaults and normalized conflict handling. A new action must not reinterpret an existing user's saved ID.

4 / TEST

Exercise the default chord, rebinding, disabled state, conflicts, wrong document kind, focus in text fields and multiple windows. Update shortcut editor discoverability and documentation.

SOURCE [src/shortcuts.rs](#) · [src/native_menu.rs](#) · [src/windowing.rs](#) · [src/app/shortcut_editor.rs](#)

How to contribute without guessing

Build the exact thing, test the contract, then inspect the pixels.

- Development setup and quality gates
- Layered UI evidence
- Performance and platform discipline

A reproducible first build

Use the repository's pinned sidecars; do not silently test against whichever Typst happens to be installed.

TERMINAL / RUN FROM THE REPOSITORY ROOT

```
git clone https://github.com/clvnkhr/tiptotyp.git
cd tiptotyp
rustc --version
cargo run --manifest-path xtask/Cargo.toml -- fetch-sidecars
cargo run -- test.typ
cargo run -- --version
```

TOOLCHAIN

Rust 1.95 / edition 2024. The documented pins are Typst 0.15.1 and Tinymist 0.15.2.

PLATFORM

Use platform build prerequisites. Poppler serves raster preview; native Wry is macOS/Windows only.

EVIDENCE

Record the real checkout's commit and tool versions. This deck is a static master-branch audit.

Baseline quality gates

xtask is outside the root workspace: check it explicitly.

TERMINAL / RUN FROM THE REPOSITORY ROOT

```
cargo fmt --all --check
cargo clippy --all-targets -- -D warnings
cargo test
cargo fmt --manifest-path xtask/Cargo.toml --check
cargo clippy --manifest-path xtask/Cargo.toml \
--all-targets -- -D warnings
cargo test --manifest-path xtask/Cargo.toml
```

BEFORE THE PATCH

Establish baseline failures and missing prerequisites before changing code.

AFTER THE PATCH

Iterate with targeted tests, then run wider checks. Check profiling-feature builds when relevant.

HONEST REPORTING

These are contributor commands. They were not executed for this source audit.

Test pyramid for this particular application

Most assertions should be deterministic and meaningful without a desktop screenshot.

LAYER	WHAT TO PROVE	EXAMPLES
Pure core / model	Transitions and invalid-state refusal.	Document keys, recovery timing, close permits, text edits.
Adapters / fake services	Correct sequencing and stale-result rejection.	Fake Tynymist, worker disconnection, delayed replies.
Filesystem / mutation	Atomic admission and conflict behavior.	Same-path saves, external modification, retained completion.
egui semantic / geometry	Widget behavior, actions and placement.	Shortcut capture, fold reveal, popup clamping.
Framebuffer snapshots	What this viewport actually painted.	Themes, text clipping, tooltip contents, selected fixture.
Native platform checks	Composition and lifecycle outside the frame-buffer.	Wry overlap, native focus, dialogs, Dock reopen, quit.

SOURCE AGENTS.md · src/save_io.rs · src/screenshot.rs · docs/ui-qa-screenshots.md

Opt-in integration evidence

Keep real-tool tests separate from fast deterministic coverage.

TERMINAL / RUN FROM THE REPOSITORY ROOT

```
TIPTOPTYP_IGNORE_SYSTEM_FONTS=1 cargo test \
persistent_watcher_compiles_errors_and_recovers \
-- --ignored

cargo bench -p tiptotyp-core --bench document

TIPTOPTYP_UI_TRACE=1 cargo run -- test.typ
```

WATCHER TEST

Real compilation, diagnostics and recovery need the expected tools. This is separate from fake-process tests.

CORE BENCH

Compare document operations under matched conditions. This does not measure native webview interaction.

GEOMETRY TRACE

Trace bounds and state without logging manuscripts. A trace line is not proof of visible behavior.

Capture the viewport you actually mean

A deterministic scene is a fixture, not an arbitrary screenshot of the current desktop.

TERMINAL / RUN FROM THE REPOSITORY ROOT

```
cargo run -- \
--ui-snapshot-scene find-replace \
--ui-theme catpuccin-latte \
--ui-screenshot-subdir screenshots/agent-review \
--ui-screenshot-exit test.typ
```

KNOW THE TARGET

Scene routing chooses a registered viewport. Check the CLI name and target in `screenshot.rs`.

KNOW THE OMISSION

An egui framebuffer cannot prove Wry composition, OS dialogs or the whole desktop.

COMPARE FAIRLY

Use fresh output and matching scene/theme/scale. The documented gallery runs a batch in one app session.

Measure the interaction—not an unrelated idle screen

Use matched optimized builds, fixtures, themes and view modes.

TERMINAL / RUN FROM THE REPOSITORY ROOT

```
cargo xtask profile --scenario large --sampler none
cargo xtask profile --scenario find \
--warmup 5 --seconds 30
cargo xtask profile --scenario tabs-switch \
--warmup 0 --seconds 2 --sampler none
cargo xtask profile --scenario hover-scroll \
--warmup 0 --seconds 2 --sampler none
```

SCENARIO SEMANTICS

Idle profiles are not typing tests. tabs-switch and hover-scroll actively exercise their named interactions.

WHAT A NUMBER MEANS

Wall spans are not CPU samples. Rebuild after changes; `--skip-build` does not prove source/binary agreement.

REGRESSION EVIDENCE

Pair measurements with work/allocation counters. Avoid flaky timing limits in ordinary correctness tests.

A pull request the next maintainer can trust

Small scope does not mean superficial evidence.

STATE THE USER EFFECT

Describe the behavior being changed, affected document types/platforms and the surface family. Include the failure case, not only the happy path.

NAME THE BOUNDARIES

Identify source identity, coordinates, persistence and operation lifetime. Explicitly say which of those your patch does not touch.

SHOW THE EVIDENCE

List commands actually run, targeted regression tests, matched captures and any native checks. Separate expected limitations from failed verification.

LEAVE THE MAP BETTER

Update the closest architecture note, source inventory or fixture. Preserve unrelated user edits and avoid opportunistic compatibility layers or sweeping cleanup.

The exact registries

Reference pages for contributors who really do want the whole list.

- All 76 shortcut actions
- All 29 menu commands
- All 43 registered snapshot scenes

All shortcut actions / 1 of 4

Exact ShortcutAction registry order. Availability and default binding are separate metadata.

REGISTRY POSITIONS	ACTION IDS	ACTION IDS
01-04	Settings / New	NewWindow / Open
05-08	OpenInNewWindow / ChangeWorkspaceRoot	Save / SaveAs
09-12	Rename / Packages	Git / StageHunk
13-16	UnstageHunk / RevertHunk	PreviousHunk / NextHunk
17-19	ExportPdf / Undo	Redo

All shortcut actions / 2 of 4

Exact ShortcutAction registry order. Availability and default binding are separate metadata.

REGISTRY POSITIONS	ACTION IDS	ACTION IDS
20–23	Cut / Copy	Paste / SelectAll
24–27	ToggleComment / Find	FindReplace / Format
28–31	SyncPreview / Problems	Explorer / Code
32–35	Split / Preview	Compile / ToggleCompilation
36–38	Complete / PointerTooltip	CaretTooltip

All shortcut actions / 3 of 4

Exact ShortcutAction registry order. Availability and default binding are separate metadata.

REGISTRY POSI- TIONS	ACTION IDS	ACTION IDS
39–42	UiScaleIn / UiScaleOut	PreviewZoomIn / PreviewZoomOut
43–46	PreviewZoomReset / Minimize	ToggleFullscreen / CloseWindow
47–50	CloseTab / NextTab	PreviousTab / CaptureUi
51–54	KeyboardShortcuts / ToggleTexMode	UseTabForPreview / ToggleFold
55–57	CollapseAll / ExpandAll	FindNext

All shortcut actions / 4 of 4

Exact ShortcutAction registry order. Availability and default binding are separate metadata.

REGISTRY POSITIONS	ACTION IDS	ACTION IDS
58–61	FindPrevious / ReplaceOne	ReplaceAll / ToggleFindCase
62–65	ToggleFindRegex / PreviewPreviousPage	PreviewNextPage / PreviewFitWidth
66–69	ExplorerSearch / RefreshWorkspace	StatusHistory / ToggleLineWrap
70–73	ToggleLineNumbers / ToggleStickyContext	UiScaleReset / WindowColor
74–76	FileMenu / EditMenu	ViewMenu

All native/in-app menu commands / 1 of 2

AppCommand is smaller than ShortcutAction. Menu metadata supplies groups and requirements.

REGISTRY POSITIONS	COMMAND IDS	COMMAND IDS
01-03	Settings / New	NewWindow
04-06	Open / OpenInNewWindow	ChangeWorkspaceRoot
07-09	Save / SaveAs	Rename
10-12	ExportPdf / CloseTab	Undo
13-15	Redo / Cut	Copy

SOURCE src/native_menu.rs

All native/in-app menu commands / 2 of 2

AppCommand is smaller than ShortcutAction. Menu metadata supplies groups and requirements.

REGISTRY POSITIONS	COMMAND IDS	COMMAND IDS
16–18	Paste / SelectAll	ToggleComment
19–21	Find / FindReplace	Format
22–24	SyncPreview / Problems	Explorer
25–27	Code / Split	Preview
28–29	Packages / Git	—

SOURCE src/native_menu.rs

All registered screenshot scenes / 1 of 3

Exact enum identifiers. CLI spelling and viewport routing are defined by the capture implementation.

REGISTRY POSITIONS	SCENE IDS	SCENE IDS
01–03	Main / Tabs	EmptyWorkspace
04–06	TabsPdf / TabsImage	StickyContext
07–09	Folding / MitexDollars	WindowColor
10–12	DelimiterMatch / RainbowBrackets	BracketSettings
13–15	FileMenu / EditMenu	SettingsWindow

SOURCE src/screenshot.rs · docs/ui-qa-screenshots.md

All registered screenshot scenes / 2 of 3

Exact enum identifiers. CLI spelling and viewport routing are defined by the capture implementation.

REGISTRY POSITIONS	SCENE IDS	SCENE IDS
16–18	SettingsColors / SettingsEditor	SettingsStatus
19–21	SettingsThemePicker / SettingsDarkThemePicker	SettingsTooltip
22–24	TypstOverridesWindow / DiagnosticTooltip	FunctionTooltip
25–27	SaveDialog / AlertDialog	OverwriteDialog
28–30	EditorContextMenu / ExplorerContextMenu	DocumentFontSelector

SOURCE src/screenshot.rs · docs/ui-qa-screenshots.md

All registered screenshot scenes / 3 of 3

Exact enum identifiers. CLI spelling and viewport routing are defined by the capture implementation.

REGISTRY POSITIONS	SCENE IDS	SCENE IDS
31–33	FontCompletion / UnicodeCompletion	SettingsFontPicker
34–36	AssetPreview / GitPanel	GitEditor
37–39	GitChunk / StatusLog	RenameDialog
40–42	WorkspaceChooser / ProblemsPanel	FindReplace
43–43	PreviewCompiling	—

SOURCE src/screenshot.rs · docs/ui-qa-screenshots.md

Ten pure-core modules: a final orientation key

This is where GUI-free policy should remain easy to exercise.

DOCUMENT / TEXT / PAIRING

Document identity and snapshots; coordinate conversion and edit batches; delimiter-pairing policy. These are not tied to native viewport widgets.

CONNECTION / PREVIEW / RECOVERY

Service state, preview artifact identity and bounded recovery. Keep time/generation inputs explicit for deterministic tests.

CLOSING / WORKFLOW

Close coordination and document-operation policy. A dirty document or pending mutation is domain state, not merely a modal being visible.

GEOMETRY / SCHEDULING

Reusable placement and work-admission policy. Extract only genuinely portable logic; keep platform effects in the outer adapters.

SOURCE [core/src/lib.rs](#) · [core/src/document.rs](#) · [core/src/text.rs](#)

What should get better—and how

Preserve the distinctive core. Remove friction before chasing every competitor.

- Evidence-backed shortcomings
- A dependency-ordered sequence of pull requests
- Acceptance gates, risks and deliberate deferrals

The diagnosis: prioritize real gaps, not fashionable rewrites

These are proposals informed by source limitations, not a claim that existing safeguards are absent.

PRIORITY	OBSERVED LIMIT / OPPORTUNITY	RECOMMENDED RESPONSE
P0 / trust and discoverability	Large evolving registries; fragmented surface implementations.	Living source inventory, action discovery and explicit lifecycle coverage.
P1 / continuity	Recent root/file restoration is not full tab/dirty-buffer recovery.	Versioned, private, opt-in restore with conflict-safe reconciliation.
P1 / project scale	Bounded index; document-only content search; single-selection tree.	Incremental indexing, cancellable project search, validated tree operations.
P1 / reading workflow	No equivalent of a dedicated detached PDF reader in the audited feature set.	Share canonical artifacts with an independent reader window.
P2 / authoring quality	No complete prose spell/count/bibliography workbench documented here.	AST-aware, offline-first assistance without disturbing Typst code.
P2 / long-term foundation	Whole-buffer editor and a substantial app adapter.	Measure first; extract seams; prototype incremental storage/layout separately.

SOURCE [README.md](#) · [editor-comparison.typ](#) · [src/project_index.rs](#) · [src/app/explorer_view.rs](#) · [src/app/settings_view.rs](#)

Do not “improve” what is already there

A credible roadmap starts by deleting incorrect diagnoses.

ALREADY PRESENT

Regex search, tabs, folding, sticky context, mi TeX projection, configurable shortcuts, hunk-level Git actions and partial-index warnings are implemented.

ALREADY PROTECTED

Versioned edits, exclusive mutations, save conflict checks, bounded preview recovery and process-wide PDF residency accounting are real existing architecture.

ACTUAL EXTENSIONS

Regex capture replacement, complete session recovery, project-wide search, multi-selection/tree moves and detached reading would extend rather than rediscover those features.

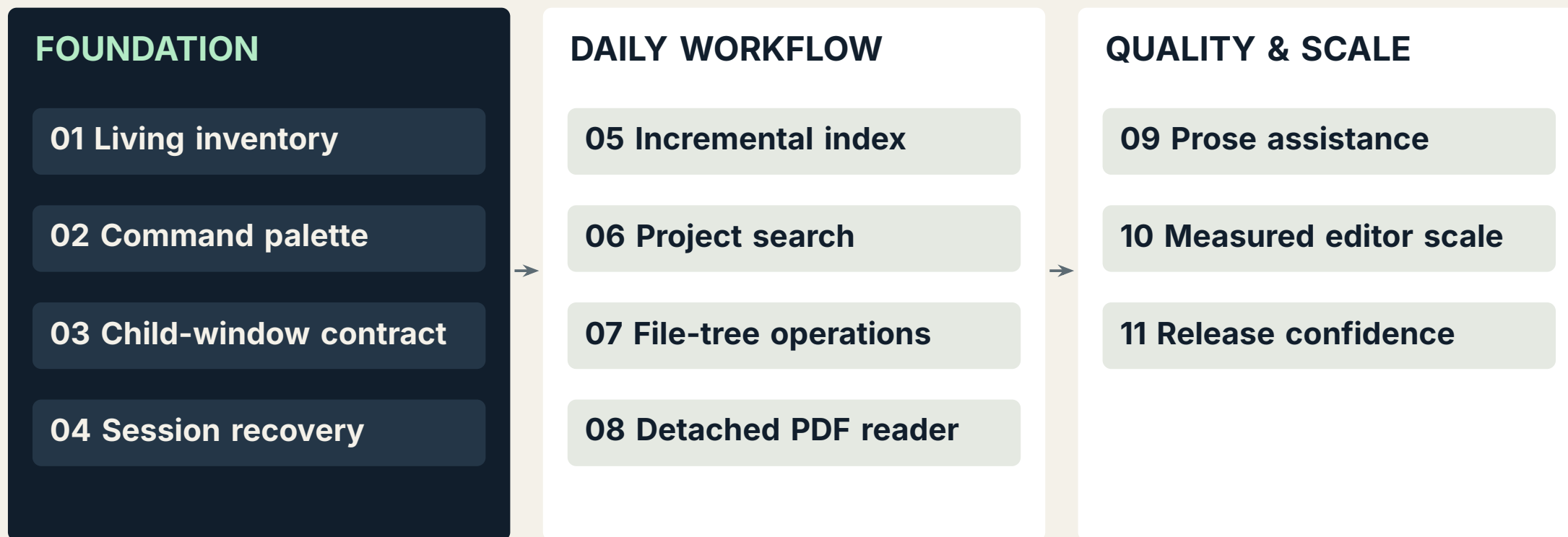
EVIDENCE DISCIPLINE

Turn each claimed gap into a reproducible user story and code check against a pinned commit. Keep competitor “not documented” separate from proven absence.

SOURCE `src/search.rs · src/app/tabs.rs · src/save_transaction.rs · src/project_index.rs · src/pdf_residency.rs`

Sequence the work so each step pays for the next

Recommended dependencies—not a calendar estimate or a promise of current implementation.



Split larger steps into small pull requests with independent state, ownership and acceptance tests.

SOURCE AGENTS.md · src/shortcuts.rs · src/app/tabs.rs · src/project_index.rs

Step 1 / make the map self-maintaining

First pull request: a pinned inventory and an evidence baseline.

CHANGE

Add a contributor architecture index that records the audited commit and derives action, popup, modal, Explorer and capture counts from their registries where practical.

TEST

Fail the inventory check when a registered item lacks documentation, ownership meta-data or an explicit “not captured” explanation. Do not hard-code branch line numbers.

TOUCH

docs/architecture/ plus a small xtask report/check. Link each surface to its owner, renderer, lifecycle, action type and capture target.

DONE WHEN

A new contributor can find one feature's entry point without opening app.rs first, and adding a registry member produces an obvious documentation task.

Step 2 / add a command palette, not another command system

Second pull request: discovery on top of existing actions.

CHANGE

Introduce searchable actions with labels, categories, current bindings and disabled reasons. Start with existing behavior; no new mutation semantics in the same patch.

TOUCH

shortcuts.rs and native_menu.rs metadata; a borrowed-input/typed-output palette renderer; owner-side dispatch in the existing application path.

TEST

Search/ranking, keyboard navigation, Escape, explicit disabled chords, document requirements, two-window focus and duplicate dispatch. Verify it does not steal ordinary typing.

DONE WHEN

Users can discover every eligible registered action and invoke it through exactly the same handler used by its keyboard/menu counterpart.

SOURCE `src/shortcuts.rs` · `src/native_menu.rs` · `src/app/completion_popup.rs`

Step 3 / finish the child-window unification

Third pull request: reduce lifecycle special cases before adding more detached surfaces.

CHANGE

Move remaining direct viewport implementations—such as Typst overrides, workspace chooser and workflow modal hosting—behind an explicit shared contract where semantics fit.

TEST

Owner closure, reopen generations, first-frame focus, blur suspension, Escape, mixed-scale placement, dormant root hosting and native-webview overlap.

TOUCH

child_view.rs, app/settings_view.rs and app/native_views.rs. Preserve deliberate differences in focus and dismissal instead of imposing one generic popup behavior.

DONE WHEN

Every transient/tool surface declares ownership, lifecycle and focus policy in one inventory; packages, shortcuts and table editor have explicit coverage or documented native checks.

SOURCE [src/child_view.rs](#) · [src/app/settings_view.rs](#) · [src/app/native_views.rs](#) · [docs/ui-qa-screenshots.md](#)

Step 4 / restore the whole session safely

Fourth pull request: continuity without silently overwriting files.

CHANGE

Persist tab order, active tab, preview entry, per-tab view state and recoverable unsaved canonical buffers in a private versioned journal. Offer restore rather than replaying writes.

TOUCH

app/tabs.rs, settings/lifecycle and a dedicated session-storage adapter. Keep durable recovery IDs distinct from in-process DocumentKey epochs.

TEST

Crash/restart, malformed journal, moved/missing file, changed disk fingerprint, duplicate path in multiple windows, miTeX mode and recovery cleanup after confirmed save.

DONE WHEN

A dirty multi-tab session can be recovered into buffers with an explicit conflict choice. No recovery action writes external files without an ordinary save transaction.

SOURCE `src/app/tabs.rs · src/settings.rs · src/app/lifecycle.rs · src/save_transaction.rs`

Step 5 / make project intelligence incremental

Fifth pull request: scale while preserving honest completeness.

CHANGE

Cache per-file parse/index results by content identity, maintain reverse dependencies and update affected entries on source/filesystem changes. Retain a configurable, visible work budget.

TEST

Import cycles, root escape/symlinks, deletions, unsaved overrides, dynamic imports, more than 256 files, rapid root switching and canceled work.

TOUCH

project_index.rs, index_jobs.rs and workspace_service.rs; expose progress and completeness through prepared Explorer data.

DONE WHEN

A one-file edit does not reparse every unchanged file; deterministic counters demonstrate the reduction; partial results remain visibly partial and stale roots never leak.

SOURCE `src/project_index.rs · src/index_jobs.rs · src/workspace_service.rs · src/app/explorer_view.rs`

Step 6 / add project search, then safe multi-file edits

Sixth pull request: search first; replacement as a separate follow-up.

CHANGE

Search file contents using a cancellable bounded worker and explicit include/exclude scope. Reuse query semantics, but return file identity plus ranges—not editor-local match state.

TEST

Unicode/regex ranges, zero-width matches, unreadable files, changed files after search, cancellation and very large trees. No filesystem scanning on the UI paint path.

TOUCH

search.rs for reusable query logic, a new project-search service/model and Explorer/panel adapters. Exclude private artifacts and expose binary/size/ignore policy.

DONE WHEN

Results navigate reliably and explain omitted work. Only then add previewed replace-across-files with per-file revalidation and explicit partial-failure reporting.

SOURCE `src/search.rs · src/workspace_service.rs · src/resource_lock.rs · src/app/explorer_view.rs`

Step 7 / finish file-tree ergonomics

Seventh pull request: multi-selection and moving files as a transaction problem.

CHANGE

Add stable multi-selection and validated moves/copies with conflict preview. Keep filesystem renaming separate from any optional rewrite of Typst imports.

TEST

Move into descendant, path collision, symlinks, partial I/O failure, dirty/open files, case-only rename and multiple windows observing the same root.

TOUCH

app/explorer_view.rs, workspace actions, workflow confirmations and exclusive-operation adapters. Update open-tab paths and relevant project indexes after successful operations.

DONE WHEN

The user sees exactly what will change, outcomes are reported per item, and failed moves do not masquerade as a successful tree-only rearrangement.

SOURCE `src/app/explorer_view.rs · src/app/workspace_view.rs · src/workflow.rs · src/worker/exclusive.rs`

Step 8 / detach the canonical PDF reader

Eighth pull request: reuse artifacts instead of starting another compiler.

CHANGE

Introduce a persistent reader window that subscribes to an accepted canonical PDF artifact. Keep independent page/zoom state and clearly mark stale output.

TOUCH

child_view/windowing hosting, PDF presentation and residency. Share immutable bytes and artifact keys; do not tie reader lifetime to whichever source tab is active.

TEST

New build, failed build, preview-entry switch, source-owner close, reader close/reopen, multi-monitor scale, page eviction and two readers sharing one artifact.

DONE WHEN

The reader follows successful output without duplicate compilation, respects process-wide page budgets and leaves export bytes unchanged. Add text/search modes as separate scoped work.

SOURCE `src/child_view.rs · src/pdf_pages.rs · src/pdf_residency.rs · core/src/preview.rs`

Step 9 / add authoring assistance that understands Typst

Ninth pull request: prose intelligence without red-squiggling every command.

CHANGE

Start with deterministic word/character statistics over clearly defined markup/prose regions. Then add optional offline spell checking with language and ignore controls.

TEST

Math, code, raw blocks, labels, references, escaped markup, multilingual text, miTeX projection and edits that invalidate ranges. Document exactly what counts as a word.

TOUCH

A source-analysis service using typst-syntax, derived-data caches and a dedicated diagnostics channel. Keep spelling separate from compiler/LSP errors.

DONE WHEN

Counts and suggestions are reproducible, do not block typing and never transmit manuscript content by default. Bibliography/Zotero workflows can follow as an optional subsystem.

SOURCE `src/editor_data.rs · src/diagnostics.rs · src/project_index.rs · editor-comparison.typ`

Step 10 / improve scale before replacing the editor

Tenth pull request series: instrument, isolate, optimize, then prototype.

FIRST

Add representative active-typing, search, fold, hover and multi-window fixtures. Record source size, view mode, warmed caches, tool versions and work counters.

NEXT

Remove measured redundant scans/allocations, isolate editor storage/layout interfaces and preserve existing undo, projection and coordinate tests.

THEN

Prototype a rope or incremental layout behind a narrow adapter. Evaluate IME composition, selection, grapheme behavior, accessibility, huge lines and native input—not only through-put.

DONE WHEN

Matched evidence shows the targeted interaction improves without correctness regressions. A storage rewrite is accepted only when it solves a measured bottleneck the smaller changes cannot.

SOURCE docs/performance.md · src/editor_data.rs · src/app/editor_view.rs · core/src/text.rs

Step 11 / turn release confidence into a maintained product

Final foundation work: platform evidence and supply-chain clarity.

PLATFORM MATRIX

Document and exercise macOS/Windows interactive preview plus Linux raster behavior. Cover focus, owner closure, dormant-host behavior and missing optional tools explicitly.

TOOLCHAIN GATES

Verify sidecar hashes, supported targets, override/fallback reporting and packaged lookup paths. Keep Typst/Tinymist compatibility upgrades in reviewable changes.

DISTRIBUTION

Replace development-only signing assumptions with the appropriate public release process where required. Test a clean machine, not only a developer shell with tools on PATH.

DONE WHEN

A release has a reproducible build identity, known capability matrix, captured regression evidence and honest unsupported/missing-tool messages. No passing screenshot stands in for native checks.

SOURCE [Cargo.toml](#) · [src/toolchain.rs](#) · [docs/architecture/0003-bundled-toolchain.md](#) · [AGENTS.md](#)

What to defer—and what would justify reconsidering it

Breadth is useful only when it serves the Typst-first purpose.

IDEA	WHY NOT MAKE IT THE FIRST PROJECT?	EVIDENCE THAT WOULD CHANGE THE ORDER
General rich-text Typst editing	Round-trip semantics are much larger than static table/miTeX transforms.	A bounded construct subset with tested source-preserving edits.
Embedded terminal	Adds process, environment and security/lifecycle scope.	Repeated contributor/user workflows that cannot be met by explicit tasks.
AI agents everywhere	Provider/auth/privacy/tool-execution complexity is not core editing reliability.	Opt-in, permissioned workflows with clear source-change review.
Real-time collaboration	Conflict resolution spans source, projection, undo and filesystem truth.	A defined collaboration model and a product need stronger than local Git.
Wholesale framework rewrite	Trades known native integration problems for an unmeasured new set.	A reproducible limitation that focused changes demonstrably cannot solve.

SOURCE editor-comparison.typ · todo.typ · src/mitex_projection.rs · AGENTS.md

The contributor compact

Build a feature. Preserve the guarantees. Leave a trail the next person can follow.

KNOW YOUR SURFACE

Which window, panel, popup or local overlay?
Which owner, lifecycle and input/focus policy?

KNOW YOUR DATA

Which document key, coordinate space, artifact generation and canonical/display representation?

KNOW YOUR EFFECT

Replaceable computation or durable mutation?
Where is admission, validation, cancellation, completion and error reporting?

KNOW YOUR PROOF

Which deterministic test, real-service check, viewport capture and native-platform observation establish that the change actually works?

SOURCE AGENTS.md · core/src/document.rs · src/child_view.rs · src/worker/exclusive.rs